

Software-as-a-Graph: Heterogeneous Graph Learning for Pre-Deployment Dependability Analysis of Asynchronous and Event-Driven Distributed Systems

Ibrahim Onuralp Yigit^{a,*}, Feza Buzluca^a

^a*Department of Computer Engineering, Istanbul Technical University, Istanbul, 34469, Turkey*

Abstract

Modern asynchronous publish-subscribe and event-driven architectures decouple components across space and time, but conceal failure paths behind message brokers and topics. Whether graph representation learning can reliably forecast cascading failure blast radii from pre-deployment Architecture-as-Code manifests without runtime telemetry remains an open empirical question. This study investigates the empirical viability and architectural boundaries of heterogeneous graph learning for static dependability analysis through Software-as-a-Graph (SaG), an experimental framework transforming manifests into typed multigraphs. We evaluate a relation-specific Heterogeneous Graph Transformer with Quality-of-Service edge encodings (HGT-QoS) against closed-form network baselines, paired with an interpretable ISO/IEC 25010/25019-grounded structural attribution layer.

Evaluation across twelve inductive scenarios and five open-source systems reveals three primary empirical boundaries. First, the learned model does not achieve a statistically significant ranking advantage over an unparameterized QoS-weighted centrality baseline ($\rho = 0.638$ vs. 0.553 , $p = 0.151$). Second, relational typing and continuous QoS encodings act as empirical substitutes rather than complements (interaction -0.199 , negative on 12/12 folds, $p = 0.0005$), providing redundant relational cues against a predominantly topological simulation target. Third, on the components that actually propagate failures every predictor loses most of its correlation, and only the typed learned model remains positive ($\rho_{>0} = +0.281$, against -0.055 to -0.092 for the training-free baselines) — a margin that establishes relational learning as the only approach retaining active-stratum signal, without being large enough to establish transfer, and one that still inverts on a synchronous RPC call tree. Computationally, while neural inference requires only 56 ms, deterministic feature extraction renders cold static analysis roughly eighteen times slower than direct discrete-event simulation (79.3s vs. 4.5s). These findings establish that unparameterized QoS-weighted centrality provides a competitive training-free baseline, that relational typing confers diminishing returns when QoS is present, and that topological feature extraction forms the principal computational bottleneck in pre-deployment static analysis.

Keywords: Heterogeneous graph neural networks, Distributed systems dependability, Publish-subscribe architecture, Cascading failures, Static system analysis, Explainable AI

1. Introduction

1.1. Motivation

Modern large-scale distributed software systems progressively employ asynchronous, event-driven, publish-subscribe (pub-sub) architectures. These architectures are widely used within autonomous driving (ROS 2 [1]),

*Corresponding author

Email addresses: yigiti@itu.edu.tr (Ibrahim Onuralp Yigit), buzluca@itu.edu.tr (Feza Buzluca)

enterprise event streams (Apache Kafka [2]), cyber-physical systems (DDS [3]), IoT deployments (MQTT [4]), cloud-native microservices [5, 6], and distributed AI/LLM serving clusters. Pub-sub architectures decouple producers and consumers across space, time, and synchronization [7]. Components interact indirectly via message topics and brokers, removing the requirement for direct static references. Contemporary middleware specifications also allow deployment-time Quality-of-Service (QoS) policies, including reliability guarantees, durability, message priorities, and delivery deadlines, to manage system performance under peak load and network stress.

While this decoupling supports elastic scalability, it also creates a substantial visibility barrier. Unlike synchronous architectures such as RESTful HTTP or gRPC, which expose interactions by explicit caller–callee paths, publish–subscribe publishers and subscribers do not maintain direct references. Consequently, chain failures, head-of-line blocking, and backpressure can propagate along concealed logical paths involving brokers, shared topics, colocated hosts, and shared libraries [8, 9]. These failures occur mainly through two mechanisms: sequential cascades, where a slow subscriber saturates a broker queue and incrementally throttles its publishers [10]; and simultaneous blast radii, where a shared library crash or host outage takes down all colocated services. Conventional architecture diagrams and static call graphs cannot represent these dissemination processes. Reducing such vulnerabilities is most effective before deployment, particularly during design and continuous integration, consistent with dependable computing principles [11, 12]. At these stages, runtime telemetry, distributed tracing, and operational logs are unavailable. Accordingly, architects and Site Reliability Engineers must identify systemically critical components, topics, and links, ascertain the root causes of their criticality, and implement focused interventions such as broker replication, decoupling over-subscribed topics, or isolating shared libraries to reduce associated risks. Here, systemic criticality denotes an entity’s tendency to serve either as an architectural single point of failure, disrupting downstream connectivity upon failure, or as an error-propagation hub capable of triggering extensive multi-hop cascading outages within the asynchronous message mesh.

These factors are also relevant to computational sustainability. Conducting architectural analysis solely on manifests eliminates the need for provisioned clusters, active containers, or live fault-injection harnesses, consequently minimizing deployment overhead and energy consumption. Nevertheless, pre-deployment static analysis does not consistently surpass simulation in speed: deterministic topological feature extraction requires 79.3s for a 520-component enterprise mesh and 239.3s for 2,000 components. By comparison, the in-process discrete-event cascade simulator completes in 0.08–4.5s (Sections 7.5.1 and 8.2). Although the learned GNN forward pass is minimal (56 ms), the overall static analysis pipeline does not provide a universal computational advantage over in-process simulation.

1.2. Problem Statement: The Architecture–Code Gap and the Black-Box AI Challenge

Pre-deployment dependability analysis in distributed architectures involves two distinct, complementary tasks. The first task is predictive forecasting: determining whether a data-driven, relation-specific graph neural network can forecast cascading failure blast radii and identify structurally critical components better than closed-form topological metrics. While closed-form metrics capture broad network connectivity, their ability to resolve multi-hop, relation-dependent cascade spread across heterogeneous channels remains an empirical question, which we test directly against such a baseline (Section 7.1). We train and evaluate the predictive pathway against independent simulation ground truth as a ranking and critical-set identification model.

Explainable Criticality Attribution (Explanation Layer) tackles the limitations of ranking alone. While a ranked shortlist identifies where risk is concentrated, it does not indicate how to remediate it. Accordingly, the predictor is paired with an interpretable structural quality profile grounded in ISO/IEC 25010 [13] and ISO/IEC 25019 [14]. This layer diagnoses structural risk archetypes grounded in standardized software quality models—distinguishing, for example, an unreplicated single point of failure from an error-propagating cascade hub or a high-coupling maintainability bottleneck—to guide targeted architectural remediations. It functions strictly as an attribution model, not a ranking model.

This separation is architectural rather than merely presentational: both pathways operate on the same graph but do not share parameters, and neither is trained on the other’s output. The coupling term that could connect them is turned off by default and reported only as an ablation (Section 4.2). This independence

allows Software-as-a-Graph (SaG) to identify components that are structurally central yet operationally low-impact, enabling a nuanced diagnosis unattainable by either pathway alone.

The distance between an architecture as designed and as realized is long-established: Perry and Wolf [15] formulated architectural erosion three decades ago, and the architectural-technical-debt literature has tracked it since [16, 17]. What we term the **Architecture–Code Gap** specializes this concept to asynchronous middleware, where the challenge is that failure semantics were never expressible in artifacts a build pipeline can inspect: *a distributed system can have pristine, bug-free source code within each service, yet remain fragile to catastrophic global outages caused by hidden single points of failure (SPOFs) or mismatched middleware Quality-of-Service (QoS) contracts*. This fragility is acute in pub-sub architectures where producers and consumers interact indirectly without static references. Classical architecture evaluations (e.g., ATAM [18, 12]) rely on manual elicitation; static code analysis [19, 20, 21, 22] inspects individual services in isolation; chaos engineering [23] requires provisioned staging clusters; and homogeneous centrality [24, 25, 26, 27] flattens typed topologies into untyped graphs. Section 2 develops each paradigm in detail.

Moreover, although machine learning has achieved significant progress in software engineering, current AI approaches to system dependability frequently operate as uninterpretable black boxes. Deep neural models generally produce scalar risk scores or latent embeddings without transparent, actionable rationales for their predictions. In mission-critical software engineering, this lack of interpretability is inadequate; developers and Site Reliability Engineers require clear explanations of component vulnerabilities and compromised architectural mechanisms to refactor code or reconfigure infrastructure effectively.

1.3. The Software-as-a-Graph (SaG) Approach

To address both the Architecture–Code Gap and the black-box AI challenge, this work puts forward **Software-as-a-Graph (SaG)**, an AI-driven pre-deployment **Static System Analysis (SSA)** framework for asynchronous and event-driven distributed systems. SaG provides the modeling and experimental foundation to investigate whether pre-deployment static representations can identify systemically critical components and attribute their structural root causes. SaG realizes Architecture-as-Code through a unified four-stage pipeline: (1) formulating the architecture as a typed, directed multigraph over Applications, Brokers, Topics, Execution Hosts, and Shared Libraries (Section 3.1); (2) projecting explicit physical connections into a QoS-weighted semantic **DEPENDS_ON** layer capturing sequential cascades and simultaneous blasts (Section 3.2); (3) training a Heterogeneous Graph Transformer (HGT) that forecasts multi-hop cascading blast radii (Section 4); and (4) evaluating an explainable, standards-grounded Reliability–Maintainability (RM) attribution profile (Section 5) to pinpoint why components are fragile.

Crucially, SaG enforces a strict **input–label independence guarantee**: learned models and attribution baselines operate exclusively on the analytical graph G_{analysis} . At the same time, ground-truth failure impacts are generated by independent discrete-event simulators operating on the raw structural topology $G_{\text{structural}}$ (Section 4.4). This separation supports the core claim by keeping prediction and evaluation independent. Section 3 formalizes the complete architectural flow and pipeline interactions (Figure 1).

Rationale for Graph Learning vs. Direct Simulation. Although discrete-event simulation $I^*(v)$ defines ground-truth criticality as well as completes in 0.08–4.5 s, it is necessary to clarify the rationale for training a graph model. From a sustainable environmental management perspective, pre-deployment static manifest analysis eliminates the substantial carbon footprint of provisioning physical staging clusters, container fleets, and live chaos-injection harnesses. However, as reported in Section 7.5.1, cold static feature extraction (79.3 s) on CPU does not outperform lightweight in-process simulation (4.5 s). In continuous integration (CI/CD) pipelines, static graph learning would achieve practical speedup under an incremental topology caching design: by caching base graph metrics across commits and extracting feature deltas only for pull-request-modified subgraphs, the sub-second neural forward pass (56 ms) would provide rapid feedback on code commits without re-running global simulation sweeps. Additionally, message passing generalizes across both labeled and unlabeled entities, enabling the scoring of entity types (such as unsimulated shared libraries or physical hosts) that a node-level simulation sweep leaves unscored. Empirical data do not support two further rationales: cascade simulation exhibits negligible stochasticity on this corpus (median test–retest 0.982), and the claim that simulation requires runnable containers is incorrect for manifest-level cascade fault injection, which

operates directly on raw manifests. Section 7.1 evaluates whether graph learning offers ranking advantages over closed-form baselines.

1.4. Research Questions

This empirical study considers five research questions:

RQ1 (Predictive Efficacy): *How accurately does heterogeneous graph learning predict cascading failure impact and identify the critical component set, compared with traditional, non-learning network indicators?*

RQ2 (Value of Architectural Typing): *Does modeling distinct entity and dependency types yield better failure predictions than homogeneous graph models on architectures the model has never seen — and does whatever advantage it confers depend on what other relational signal the model already has?*

RQ3 (QoS Encoding and Robustness): *(i) Do middleware Quality-of-Service contracts carry signal a purely structural score discards, (ii) does that signal compose with or substitute for architectural typing, (iii) do the framework’s simulation oracles agree with one another, and (iv) are the reported orderings robust to the free parameters of the scorer and of the ground truth?*

RQ4 (Real-World Generalization): *How effectively does the framework transfer zero-shot to authentic, real-world distributed systems across autonomous driving (ROS 2), cloud-native microservices, smart home IoT, and industrial edge computing?*

RQ5 (Analysis Cost): *What does pre-deployment analysis cost at CI/CD time, which pipeline stage dominates that footprint, and how does it compare against the discrete-event simulation it is intended to displace?*

1.5. Key Contributions

This paper makes four principal contributions:

- 1. Heterogeneous Graph Learning for Dependability, and Its Boundaries:** A relation-specific Heterogeneous Graph Transformer that forecasts cascading blast radii from Architecture-as-Code manifests, incorporating 16-D QoS edge encodings and an auxiliary component reliability head (Section 4). Ablated separately under inductive distribution shift across twelve architectures, relation-specific HGT demonstrates consistent out-of-distribution generalization ($\Delta\rho = +0.234$ over an untyped baseline, $p = 0.0005$). Crucially, typing and continuous QoS encodings act as empirical substitutes rather than complements. Against an unparameterized QoS-weighted centrality baseline, learned ranking does not establish a statistically significant advantage ($+0.085$, $p = 0.151$). We report this substitution effect and the empirical boundary as primary findings (Sections 7.1–7.2).
- 2. A Formal Typed Architecture Model:** A multigraph representation that derives logical dependencies from physical pub-sub linkages and distinguishes sequential cascade propagation from simultaneous multi-consumer library failures (Section 3).
- 3. A Standards-Grounded Explanation Layer (a design contribution):** An interpretable Reliability–Maintainability model based on ISO/IEC 25010/25019 that distinguishes single-point-of-failure exposure from error-propagation reach (Section 5). Its evidential status is explicitly bounded: it achieves modest ranking performance ($\rho = 0.205$, below unweighted centrality), degree centrality outperforms it on pooled detection, and its elicited AHP weights are anti-predictive against a uniform prior. It presents a structural template for qualitative attribution (Section 8.4).
- 4. Empirical Benchmark, Real-World Transfer, and Cost Profile:** A study across seventeen system architectures totaling 2,812 components: twelve synthetic topologies (2,461 components across LOSO folds) and five open-source reference systems (351 components) under strict graph-view separation. We characterize pipeline costs, showing that the neural model accounts for only 0.02% of runtime, whereas deterministic feature extraction dominates (79.3s vs. 4.5s for simulation), refuting the premise that static analysis is inherently faster than in-process simulation (Sections 6–7).

Relationship to the authors' prior work. A previous conference paper [28] introduced the preliminary multigraph formulation and deterministic quality model on synthetic topologies. This JSS manuscript substantially extends that work through incorporating the complete predictive HGT pathway with 16-dimensional QoS edge encoding (Section 4); inductive LOSO cross-validation (Section 7.2); zero-shot evaluation throughout five open-source systems (Section 7.4); empirical cost and sustainability characterization (Section 7.5); multi-oracle convergent validity and graph-view separation (Sections 4.3–4.4); and system-wide sensitivity analyses (Section 7.3). We retained formalisms from the conference paper only in restructured portions of Sections 3 and 5.

1.6. Paper Organization

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 formalizes the SaG multigraph model and dependency projections. Section 4 details the Heterogeneous Graph Transformer and simulation oracles. Section 5 presents the ISO/IEC-grounded explanation layer. Section 6 outlines the experimental protocol, and Section 7 reports empirical results for RQ1–RQ5. Section 8 discusses practical consequences, sustainability, threats to validity, and limitations. Section 9 concludes.

2. Related Work

The central claim of this study is that joining four core research domains (1) dependability, performance, and sustainability in distributed software systems; (2) static code and system analysis; (3) software quality analysis and multi-criteria evaluation; and (4) graph representation learning and explainable artificial intelligence (XAI)—enables prediction of systemic cascading vulnerabilities and performance degradation from Architecture-as-Code descriptors before runtime infrastructure provisioning.

2.1. Dependability, Performance, and Sustainability in Distributed Software Systems

The publish–subscribe (pub-sub) and asynchronous event-driven paradigms decouple communicating entities in space, time, and synchronization, enabling elastic scalability and high throughput [7]. Contemporary middleware standards, including ROS 2 [1], Apache Kafka [2], DDS [3], and MQTT [4], regulate these exchanges through fine-grained Quality-of-Service (QoS) policies. In cloud-native microservice meshes and distributed AI/LLM serving infrastructures, asynchronous message passing and queueing topologies form the primary communication substrate and influence tail latencies, throughput bottlenecks, and hardware utilization.

Previous research on dependability and performance has primarily emphasized runtime mechanisms, such as dynamic consensus, broker clustering, adaptive backpressure throttling, autoscaling, and automated failover. Concurrently, chaos engineering and runtime fault injection [23] introduce disruptions into live staging or production clusters to observe degradation and recovery. Although runtime testing provides realistic operational validation, it requires fully provisioned clusters, risks service disruption, and incurs substantial compute overhead. This limits its utility during early architectural design and lightweight commit-level continuous integration (CI/CD), situating such workflows among development-time computations scrutinized in green software engineering [29, 30, 31, 32, 33]. In contrast, this study investigates pre-deployment dependability analysis directly from Architecture-as-Code descriptors before provisioning runtime infrastructure, eliminating cluster provisioning overhead and preempting production restart storms.

Predicting dependability from architectural descriptions has a rich analytical lineage. Cheung’s absorbing-Markov-chain model [34] derives system reliability from component reliabilities and transfer-of-control graphs; Goseva-Popstojanova and Trivedi [35] systematize subsequent state-based, path-based, and additive families, as Immonen and Niemelä [36] survey. Model-driven frameworks, such as the Palladio Component Model [37] and layered queueing networks [38], predict performance and reliability from parameterized component specifications, while annotation-based approaches such as the AADL Error Model Annex [39] generate fault trees from declared error states. These methods address broader questions than those considered in this study, but require pre-calibrated operational profiles and failure rates unavailable at commit time. In contrast, SaG

asks a single targeted question from manifests alone: which components’ failures propagate furthest through the declared topology?

A substantial body of literature localizes faults in microservices using operational telemetry: Seer [40] and Sage [41] predict QoS violations from hardware counters and traces; MicroRCA [42] and TraceRCA [43] isolate root causes over service-dependency graphs; and DeepTraLog [44], Eadro [45], and MicroCause [46] apply graph neural networks to multimodal traces, logs, and metrics (surveyed across 98 papers by Zhang et al. [47]). In an industrial benchmark study, Zhou et al. [48] show that cascading outages in synchronous microservices result from thread-pool exhaustion, RPC timeouts, and recursive retry storms propagating along call trees. In contrast, pub-sub failures propagate via broker queue saturation and message starvation. All these approaches require a running cluster emitting runtime telemetry, whereas SaG addresses the pre-deployment complement by operating on static manifests before code execution.

2.2. Static Code Analysis (SCA) vs. Static System Analysis (SSA)

Traditional **Static Code Analysis (SCA)** tools (e.g., SonarQube [19]) inspect source code Abstract Syntax Trees (ASTs) within individual services. They evaluate cyclomatic complexity [20], class cohesion, module coupling (e.g., Lack of Cohesion in Methods [LCOM], Coupling Between Objects [CBO]) [21, 22], and code duplication to flag internal code smells and defect-prone modules [49, 50, 51, 52]. However, SCA cannot observe runtime communication topology: it does not capture inter-service messaging channels, message broker queue saturation, or cross-host failure propagation.

Static recovery of system-level structure is an active area of complementary research. Bushong et al. [53] derive communication diagrams from static code analysis, and a recent review evaluates nine architecture recovery tools [54]. In robotics and publish-subscribe ecosystems, specialized static analysis tools such as HAROS [55] extract computation models to detect structural defects prior to system launch. This body of literature aims to construct accurate representations of implemented systems to facilitate comprehension and drift detection. In contrast, SaG uses a declared topology as input to forecast cascade blast radii.

To bridge the “Architecture–Code Gap,” **Static System Analysis (SSA)** extends static analysis from single-service source code to the global system architecture. Through modeling distributed applications, message topics, brokers, execution nodes, and shared libraries as a connected multigraph, SSA propagates code-level quality metrics across architectural dependencies. This approach enables engineering teams to detect structural anti-patterns [56, 57] and architectural technical debt [58] early during continuous integration (CI/CD) [59, 60], before defective topologies enter production.

2.3. Software Quality Models and Multi-Criteria Evaluation

Software product quality is standardized by the **ISO/IEC 25010:2023** product quality model [13] and the **ISO/IEC 25019:2023** Quality-in-Use model [14]. ISO/IEC 25010:2023 defines three closely intertwined characteristics: **Reliability** (comprising Faultlessness, Availability, Fault Tolerance, and Recoverability), **Maintainability** (Modularity, Reusability, Analyzability, Modifiability, and Testability), and **Efficiency of Performance** (Time Behavior, Resource Utilization, and Capacity). SaG operationalizes a strict subset derivable from deployment topology: Availability and Fault Tolerance under Reliability, and Modularity, Modifiability, and Analyzability under Maintainability (Section 5.1). The characteristics of Faultlessness, Recoverability, Reusability, and Testability remain outside the scope of topological analysis.

Software engineering measurement explicitly distinguishes between *internal quality* (inbuilt structural attributes assessed on static artifacts at rest) and *external quality* (runtime dependability and behavioral characteristics noted during system execution) [61, 62]. In distributed architectures, architectural technical debt, such as over-centralized message topics or unreplicated brokers, degrades internal quality and can precipitate severe external performance bottlenecks, queue congestion, and outages.

Aggregating multi-attribute structural metrics into an auditable quality score presents a classic Multi-Criteria Decision Making (MCDM) challenge. The Analytic Hierarchy Process (AHP) [63] supplies a structured pairwise-comparison methodology with an explicit Consistency Ratio ($CR \leq 0.10$) to guarantee consistency among elicited judgments. While this statistic detects inconsistency, it does not identify matrices completed from predetermined answers, a limitation addressed for the weights in this study (Section 5.2).

This work applies AHP to establish an audited, explainable Reliability–Maintainability (RM) quality baseline together with learned graph models.

2.4. Graph Representation Learning and Explainable AI

Network science offers established centrality indices for identifying critical nodes, such as degree, closeness, betweenness centrality [24, 26], articulation points, and PageRank [25, 27]. Fundamental studies on network robustness [10], cascading overloads [8], and interdependent networks [9] model disruption propagation over interconnected topologies. While percolation models provide natural comparators, this study’s training-free baselines are centrality-based (Section 6.2). We propose targeted percolation fragmentation as a route for future baseline comparisons.

Standard network measures present two major shortcomings in the context of software architectures. First, Dimensional Collapse arises when a single centrality value does not distinguish the underlying reasons for a component’s criticality, such as differentiating between an isolated single point of failure, an error-propagating cascade hub, or an over-shared library. Second, Semantic Collapse occurs when unweighted metrics treat all nodes and edges equivalently, conflating fundamentally distinct architectural entities, such as asynchronous message topics, shared libraries, and physical execution hosts.

To tackle the limitations of hand-engineered metrics, recent research applies machine learning to network vulnerability analysis (e.g., FINDER [64], DrBC [65], PowerGraph [66]). However, most models employ homogeneous message passing (GCN [67], GraphSAGE [68], GAT [69]) and indiscriminately average signals across connection types. Given the inherent heterogeneity of distributed software architectures, such models obscure entity boundaries and fail to generalize to out-of-distribution scenarios. Heterogeneous Graph Neural Networks (RGCN [70], HAN [71], HGT [72], MAGNN [73]) address this issue through relation-specific transformations. The present study uses the Heterogeneous Graph Transformer (HGT) [72] to preserve typed relational semantics when forecasting cascade blast radii. Graph learning has also been directly applied to microservice topologies; for instance, Khodabandeh et al. [74] predict future service interactions using graph attention over temporally segmented call graphs. However, that approach forecasts edge existence relying on observed interaction history. In contrast, the current method accepts a declared topology as input and predicts the blast radius resulting from node removal.

A major challenge in applying modern artificial intelligence to software engineering is the black-box barrier: deep neural models generate risk scores or continuous embeddings without elucidating basic structural causality. In production environments, these uninterpretable risk rankings hinder actionable decision-making, as developers and site reliability engineers (SREs) cannot determine whether to replicate hosts, configure circuit breakers, or refactor shared libraries.

Existing graph neural network (GNN) explanatory methods, such as GNNExplainer [75] and PGExplainer [76], identify influential subgraphs through edge masking or parameterized learning. Although valuable, these approaches interpret models using internal hidden representations rather than standardized software engineering concepts. SaG tackles this drawback through a decoupled dual-pathway design. The predictive HGT pathway reveals typed mutual-attention distributions that indicate which architectural relations propagated the cascade (Section 7.3.3). At the same time, the deterministic explanation layer attributes fragility to standardized ISO/IEC quality sub-characteristics (Section 5), translating raw predictions into implementable, cost-effective remediations.

3. The Software-as-a-Graph (SaG) Architectural Model

Figure 1 presents the end-to-end architecture of the SaG framework. The shared front end processes Architecture-as-Code manifests, constructs a typed multigraph, projects implicit runtime interactions throughout a QoS-weighted logical dependency layer, and extracts typed node properties. These features feed two independent pathways with no shared parameters. The predictive pathway (Section 4) forecasts cascading failure blast radii using a Heterogeneous Graph Transformer. The explanation layer (Section 5) decomposes fragility into Reliability and Maintainability quality profiles.

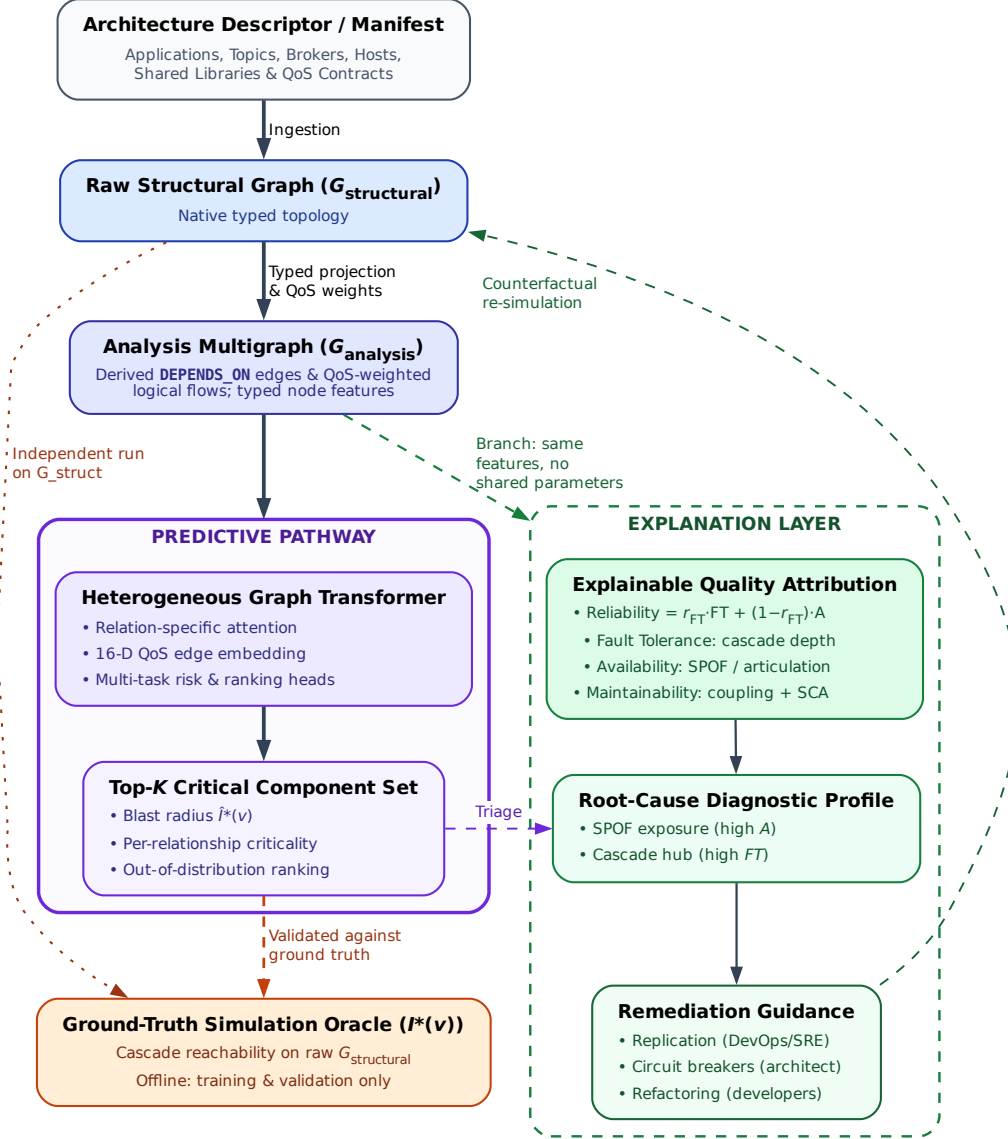


Figure 1: End-to-end architecture of the SaG framework. The central assertion is that the predictive pathway (Section 4) forms the primary sequence: manifest ingestion $\rightarrow$ typed multigraph $\rightarrow$ QoS-weighted **DEPENDS_ON** projection $\rightarrow$ typed node properties $\rightarrow$ heterogeneous graph learning $\rightarrow$ a ranked critical set (with an architectural interface for per-relationship criticality reserved for future work) $\rightarrow$ the ground-truth simulation oracle (Section 4.3) that evaluates it. The oracle completes the predictive pathway’s training loop and operates solely on $G_{structural}$; it functions strictly offline and is excluded from inference (Section 4.4), as indicated by the dashed edge. The explanation layer (Section 5) differs from this sequence: it re-enters from the analysis multigraph, generates a standards-based quality profile from the same typed features without sharing parameters with the predictor, and is accessed through triage instead of direct data flow.

This section formalizes the Software-as-a-Graph multigraph representation (Section 3.1), the QoS-aware weighting and logical dependency derivation rules (Section 3.2), the dual graph views (Section 3.3), and the typed node feature encodings (Section 3.4).

3.1. Formal Multigraph Definition

A complex distributed software system is formally represented as a typed, weighted, directed multigraph:

$$\mathcal{G} = (V, E, \tau_V, \tau_E, w_V, w_E) \quad (1)$$

where:

- V is the set of system entities, partitioned into five disjoint categories $\mathcal{T}_V = \{\text{app, broker, topic, host, lib}\}$ such that $V = \bigcup_{t \in \mathcal{T}_V} V_t$:

$$V = V_{\text{app}} \cup V_{\text{broker}} \cup V_{\text{topic}} \cup V_{\text{host}} \cup V_{\text{lib}} \quad (2)$$

To prevent conflation of graph vertices with physical compute machines, V_{host} is designated as *Execution Hosts* (physical hosts or virtualized execution nodes).

- E is the set of directed edges connecting entities.
- $\tau_V : V \rightarrow \mathcal{T}_V$ and $\tau_E : E \rightarrow \mathcal{T}_E$ are typing functions assigning entity and relationship categories.
- $w_V : V \rightarrow (0, 1]$ and $w_E : E \rightarrow (0, 1]$ are weighting functions representing entity criticality and connection strength. For applications and shared libraries, $w_V(v)$ is initialized from static code metrics as $w_V(v) = 1 - \text{CQP}(v)$ (where CQP denotes the Code Quality Penalty, Section 3.4). It defaults to 1.0 when static code metrics are absent or for infrastructure entities ($w_V(\text{host}) = 1.0$).

Table 1 summarizes the five entity types and six structural edge types defined in the SaG model, together with their semantics and representative distributed-system implementations.

Table 1: Entity and structural edge types in the SaG model.

Entity Type ($\mathcal{T}_V$)	Architectural Role	Concrete System Examples
Application (V_{app})	Autonomous process producing/consuming messages	ROS 2 node, Kafka microservice, MQTT client
Broker (V_{broker})	Message routing and queuing intermediary	RabbitMQ exchange, Mosquitto, EMQX broker
Topic (V_{topic})	Named logical communication channel	/sensor/lidar, orders.payment.completed
Execution Host (V_{host})	Physical host or virtualized execution environment	Bare-metal server, Kubernetes worker, Cloud VM
Library (V_{lib})	Shared software package or runtime dependency	Kafka client, OpenCV, Protobuf runtime
Structural Edge ($\mathcal{T}_E$)	Direction	Semantic Meaning
PUBLISHES_TO	App/Library $\rightarrow$ Topic	Component publishes messages to topic
SUBSCRIBES_TO	App/Library $\rightarrow$ Topic	Component consumes messages from topic
ROUTES	Broker $\rightarrow$ Topic	Broker manages and routes topic traffic
RUNS_ON	App/Broker $\rightarrow$ Host	Process is hosted on physical/virtual host
CONNECTS_TO	Host $\rightarrow$ Host	Physical network link between hosts
USES	App $\rightarrow$ Library	Application links to shared library dependency

Application and Library entities also incorporate static code metrics generated by Static Code Analysis (SCA) tools: lines of code, cyclomatic complexity, coupling between objects, and lack of cohesion in methods. These metrics link code-level fragility directly to topological analysis.

Notation. Entity and edge types: Table 1. Simulation oracles: Table 3.

$G_{\text{structural}}$	Raw multigraph; oracles only	$Q(v)$	RM composite quality score
G_{analysis}	DEPENDS_ON projection; predictor input	ρ	Spearman ρ , full population
V_{app}	Application nodes; the scored population	$\rho_{>0}$	Spearman ρ , active stratum ($I^* > 0$)
$w(t), w(e)$	QoS topic weight, edge weight	$F_1@K$	Critical-set overlap, $K = 0.20 V_{\text{app}} $
$I^*(v)$	Primary cascade-reachability oracle	$I_{\text{comp}}, I_{\text{dyn}}, I_M$	Further oracles (Table 3)

3.2. QoS-Aware Weights and Logical Dependency Derivation

In distributed middleware, communication links differ in strength according to their Quality-of-Service (QoS) contracts. For example, a **RELIABLE** topic with **TRANSIENT_LOCAL** durability creates a stronger binding between communicating services than a **BEST_EFFORT** telemetry stream.

Each topic t carries an intrinsic criticality weight $w(t) \in (0, 1]$ combining its declared QoS semantics with two runtime-stress modulators: payload size and publication frequency:

$$w(t) = \alpha_{\text{top}} \cdot \text{QoS}(t) + \beta_{\text{top}} \cdot \text{SizeNorm}(t) + \gamma_{\text{top}} \cdot \text{FreqNorm}(t), \quad (\alpha_{\text{top}}, \beta_{\text{top}}, \gamma_{\text{top}}) = (0.75, 0.15, 0.10) \quad (3)$$

where the QoS term is an AHP-weighted aggregate of the declared contract:

$$\text{QoS}(t) = w_{\text{rel}} \cdot q_{\text{rel}} + w_{\text{dur}} \cdot q_{\text{dur}} + w_{\text{prio}} \cdot q_{\text{prio}}, \quad (w_{\text{rel}}, w_{\text{dur}}, w_{\text{prio}}) = (0.24, 0.62, 0.14) \quad (4)$$

Here, $q_{\text{rel}}, q_{\text{dur}}, q_{\text{prio}} \in [0, 1]$ represent normalized reliability, durability, and transport-priority scores mapped from manifest policies: $q_{\text{rel}} \in \{0.0, 1.0\}$ (best-effort vs. reliable), $q_{\text{dur}} \in \{0.0, 0.5, 1.0\}$ (volatile, transient-local, persistent), and $q_{\text{prio}} \in \{0.0, 0.5, 1.0\}$ (low, medium, high). Durability carries highest weight as it determines state preservation across restarts. The sub-weights derive from a Saaty pairwise matrix with consistency ratio $CR = 0.016 \leq 0.10$.

The modulators $\text{SizeNorm}(t)$ and $\text{FreqNorm}(t)$ are logarithmically compressed and clamped to $[0, 1]$:

$$\text{SizeNorm}(t) = \min \left(1.0, \frac{\log_2(1 + B(t))}{20} \right), \quad \text{FreqNorm}(t) = \min \left(1.0, \frac{\log_{10}(1 + F(t))}{3} \right) \quad (5)$$

Here, $B(t)$ denotes the message payload size in bytes (with a 1 MiB design envelope, which represents the practical DDS sample ceiling before RTPS fragmentation becomes dominant), and $F(t)$ represents the nominal publication frequency in Hertz. The final weight $w(t)$ is clamped to $[0.01, 1]$ to ensure that best-effort edges remain visible during graph traversals. Each structural communication edge incident on t (**PUBLISHES_TO**, **SUBSCRIBES_TO**, **ROUTES**) inherits $w_E(e) = w(t)$ together with the topic's QoS vector.

The outer split $(\alpha_{\text{top}}, \beta_{\text{top}}, \gamma_{\text{top}})$ is a convex combination. Sweeping the full simplex leaves induced orderings intact ($\rho \geq 0.919$), moving downstream ranking by only 0.031 (**Topo-QoS**) and 0.007 (RM). Under Morris screening none of the three constants is load-bearing ($\mu^* \leq 0.025$). The split is therefore a documented convention rather than a sensitive parameter.

3.2.1. Logical Dependency Projection (**DEPENDS_ON**)

Structural edges represent explicit deployment connections but do not capture implicit runtime dependencies. For instance, a subscriber depends on a publisher, yet no direct edge connects them in publish-subscribe architectures. To address this, a single unified semantic relation, **DEPENDS_ON**, is derived and directed from *dependent* to *dependency* ("if target fails, source is impacted"), following the six projection rules detailed in Table 2. The resulting weight $w \in (0, 1]$ quantifies the magnitude of operational coupling and indicates the conditional likelihood that a disruption in the dependency propagates to the dependent.

Table 2: The six **DEPENDS_ON** logical dependency projection rules.

Rule	Dependency Category	Structural Pattern (Dependent $\rightarrow$ Dependency)	Derived Weight (w)
1	app_to_app	Subscriber $\rightarrow$ Publisher (via shared Topic, incl. transitive USES)	$1 - \prod_{t \in T} (1 - w(t))$
2	app_to_broker	Publisher/Subscriber $\rightarrow$ Broker routing its topics	$1 - \prod_{t \in T} (1 - w(t))$
3	host_to_host	Host $\rightarrow$ Host (lifted from inter-host app dependencies)	$\max_{u \in \text{hosted}(h_1), v \in \text{hosted}(h_2)} w_{\text{DEPENDS_ON}}(u \rightarrow v)$
4	host_to_broker	Host $\rightarrow$ Broker (lifted from hosted app dependencies)	$\max_{u \in \text{hosted}(h)} w_{\text{DEPENDS_ON}}(u \rightarrow b)$
5	app_to_lib	Application $\rightarrow$ Shared Library it USES	$H(w_V(\text{app}), w_V(\text{lib}))$
6	broker_to_broker	Broker $\leftrightarrow$ Broker (shared physical fault-domain colocation, symmetric)	$w_V(\text{host})$

Rules 1 and 2 aggregate the set of topics T connecting a component pair using a probabilistic union rather than a maximum [77, 78, 79]. This approach confirms that additional parallel failure vectors increase coupling monotonically while maintaining $w \in (0, 1]$. Rule 5 employs the harmonic mean $H(x, y) = 2xy/(x + y)$ [80] to combine the vertex weights of the consuming Application and the shared Library, consequently balancing caller and dependency criticality. Rules 3 and 4 extend application-level dependencies across host boundaries using the maximum coupling weight.

3.2.2. Sequential Cascades vs. Simultaneous Blasts

A central principle of the SaG model is the distinction between two degradation modes. In a **Sequential Cascade (Rule 1)**, a failed publisher starves downstream subscribers sequentially through message queues and topic buffers. In a **Simultaneous Blast (Rule 5)**, a crashed library or execution host causes all consuming applications and colocated brokers to fail instantaneously in a single shared-fate event. Preserving entity types and relation-specific projection rules lets the SaG model represent both mechanisms, whereas untyped homogeneous graphs collapse them into indistinguishable edges.

Rule 6 is the only symmetric rule: two brokers colocated on the same host share that host’s physical failure domain ($w = w_V(\text{host})$). Colocated brokers compete for resources, and host crashes halt them simultaneously. Rule 6 applies in four of eight detection benchmark contexts (12 directed edges). Because simulation oracles operate strictly on $G_{\text{structural}}$ (Section 4.4), Rule 6 has zero influence on ground-truth failure labels.

3.3. Dual Graph Views and Architectural Layers

The SaG framework consists of two distinct representations. The **Structural Graph** ($G_{\text{structural}}$) is the raw deployment graph containing physical and structural relations (PUBLISHES_TO, ROUTES, RUNS_ON, USES); it preserves the untransformed deployment topology. The **Analysis Graph** (G_{analysis}) is the projected graph containing derived DEPENDS_ON edges annotated with QoS weights and ingested SCA metrics. We compute all GNN embeddings and analytical metrics on the analysis graph (Figure 2).

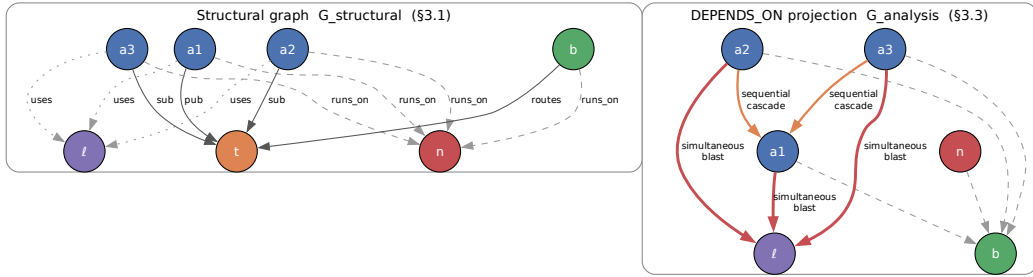


Figure 2: Running example: the raw structural graph (left) and the **DEPENDS_ON** projection derived from it (right). The projection makes implicit runtime dependencies explicit—a subscriber depends on the publishers of its topics even though no structural edge joins them—while the simulators continue to operate on the structural view alone.

G_{analysis} is further organized into four analytical layers (Application, Middleware, Infrastructure, and Global System), enabling criticality evaluation at subsystem levels consistent with hierarchical frameworks such as MIL-STD-498 [81].

3.4. Typed Node Feature Encoding

Within the SaG architecture, both the predictive pathway (Section 4) and the explanation layer (Section 5) utilize the same unified typed node properties from G_{analysis} . The predictive pathway projects these properties per entity type before heterogeneous message passing, while the explanation layer aggregates them into its quality profile. All five entity types share indices 0–17, an 18-dimensional block of topological metrics. The block comprises PageRank (0), reverse PageRank (1), betweenness centrality (2), closeness centrality (3), eigenvector centrality (4), in-degree and out-degree centralities (5–6), clustering coefficient (7), undirected articulation score (8), bridge ratio (9), total, in-bound, and out-bound QoS weights (10–12), multi-path connectivity index (13), path complexity (14), fan-out criticality (15), directed articulation score (16), and the Connectivity Degradation Index (CDI, index 17). CDI is the single most expensive metric in the block and dominates the deterministic analysis cost characterized in Section 7.5. Because it is a predictor input and not only a term of the Availability score (Section 5.2), it cannot be gated away without changing both pathways. The deterministic analysis stage produces these metrics, and Section 7.5 describes its computationally demanding cost. All topological metrics in this block are normalized to $[0, 1]$ within each graph: degrees are normalized by $|V| - 1$, betweenness and closeness follow standard network formulations, and reverse PageRank is normalized to unit sum. This normalization prevents raw graph size and component counts

from controlling multi-layer perceptron projections during cross-scenario inductive transfer. Type-specific blocks extend the feature set to between 19 and 25 dimensions. These add source-code metrics and the Code Quality Penalty for Applications, two reverse-USES blast-radius drivers for Libraries, queue capacity for Brokers, publisher/subscriber counts and ordinal QoS criticality for Topics, and CPU and memory allocation for Execution Hosts (V_{host}).

The shared block includes the graph composition, which is relevant to understanding Section 7: betweenness, closeness, reverse PageRank, and articulation score are topological summaries computed before model evaluation. Consequently, a learned model is not the only way to derive a criticality score from structure, which ensures that the closed-form baselines serve as fair comparators rather than strawman alternatives.

4. Graph Learning for Failure-Impact Prediction

Cascading failure impact in distributed software systems indicates non-linear, multi-hop, and relation-dependent characteristics. Outages propagate through architectural relations and dependencies beyond immediate neighbors. Whether a closed-form combination of standard centrality indices can adequately capture these complicated dynamics remains an open empirical question. Consequently, the primary predictive approach described in Section 1.2 utilizes a learned graph model, which is evaluated in Section 7.1 against a closed-form baseline. The learned model does not manifest significant improvement over the baseline in out-of-distribution ranking.

This section presents the Heterogeneous Graph Transformer (HGT) architecture and its typed edge encodings (Section 4.1), the multi-task prediction heads and dimension-masked loss formulation (Section 4.2), the ground-truth simulation oracles (Section 4.3), and the input-label independence guarantee designed to prevent data leakage (Section 4.4).

4.1. Heterogeneous Graph Transformer Architecture

Distributed systems comprise heterogeneous entity types (Applications, Libraries, Brokers, Topics, Execution Hosts) and diverse interaction semantics (PUBLISHES_TO, SUBSCRIBES_TO, ROUTES, RUNS_ON, CONNECTS_TO, USES, DEPENDS_ON). We therefore employ a three-layer **Heterogeneous Graph Transformer (HGT)** architecture [72], implemented within PyTorch Geometric [82], with hidden dimension $D = 64$ and $H = 4$ attention heads. This architecture guarantees that typed relations, rather than simple adjacency, govern failure-impact forecasting.

4.1.1. Continuous-Categorical Edge Feature Encoding (16-D)

To capture continuous QoS constraints and channel semantics, SaG encodes each directed edge $e = (u, v)$ as a 16-dimensional continuous-categorical vector $e_{uv} \in \mathbb{R}^{16}$. Index 0 is the scalar coupling weight $w_E(e) \in (0, 1]$ of Section 3.2; index 1 is the normalized count of simple paths through e ; indices 2–8 one-hot encode the seven structural and derived relations; and indices 9–15 carry middleware QoS parameters on PUBLISHES_TO and SUBSCRIBES_TO edges, zeroed elsewhere. Six QoS dimensions are active in our corpus — reliability, durability, message priority, a heterogeneity flag raised when an edge’s QoS triple departs from its scenario’s modal profile, and the deadline pair (an active flag and $\log_{10}(1 + \text{deadline_ns}/10^6)$, populated on 463 of 615 topics 75%). The seventh, $\log_{10}(1 + \text{max_blocking_ms})$, is a schema provision for hard real-time DDS and ROS 2 profiles and is zero throughout.

An edge projection module maps e_{uv} into the hidden space: $e'_{uv} = W_{\text{edge}}e_{uv}$. Before relational attention computation, the current projection vector is incorporated directly into the target node representation: $\tilde{h}_v = h_v + e'_{uv}$.

4.1.2. Type-Specific Projection and Heterogeneous Message Passing

For each source node u and target node v connected by meta-relation $\tau(e) = (\tau(u), \phi(e), \tau(v))$, SaG follows the Heterogeneous Graph Transformer formulation of Hu et al. [72] implemented via PyTorch Geometric’s HGTConv [82]. Entity-specific projections $W_{\tau(v)}$ first map raw features $x_v \in \mathbb{R}^{19-25}$ into the shared D -dimensional hidden space: $h_v^{(0)} = \text{LayerNorm}(\text{GELU}(W_{\tau(v)}x_v))$. Relational mutual attention

across H heads incorporates type-parameterized Key ($K(u) = h_u^{(l-1)} W_K^{\tau(u)}$), Query ($Q(v) = \tilde{h}_v^{(l-1)} W_Q^{\tau(v)}$), and Value ($V(u) = h_u^{(l-1)} W_V^{\tau(u)}$) projections along with the edge representation $\tilde{h}_v = h_v + e'_{uv}$. Crucially, attention scores scale by a learned per-meta-relation prior $\mu_{\langle \tau(u), \phi(e), \tau(v) \rangle}$ (`p_rel` in PyG), which lets the model weight an entire relation triple up or down independently of individual node embeddings; this parameter directly captures the relational typing effect evaluated in Section 7.2. Message passing operates bidirectionally across both forward and transposed relation views (G_{analysis} and $G_{\text{analysis}}^\top$) to capture downstream starvation and upstream backpressure simultaneously, followed by residual aggregation, dropout ($p = 0.10$), and layer normalization across layers $l \in \{1, \dots, L\}$.

Training Protocol and Optimization Hyperparameters. Models are optimized end-to-end using AdamW with an initial learning rate of $\eta = 3 \times 10^{-4}$, weight decay of 10^{-4} , and dropout probability $p = 0.10$ applied post-attention. Learning rates follow a cosine decay schedule with warm restarts (CosineAnnealingWarmRestarts, $T_0 = 75$, $T_{\text{mult}} = 2$, $\eta_{\text{min}} = 3 \times 10^{-6}$). Training runs for up to 300 epochs, with early stopping set to 30 epochs based on validation loss over labeled nodes. Inductive subgraphs are processed per scenario using full-graph inductive packing without mini-batch subsampling, and validation masks isolate held-out nodes to prevent information leakage across data splits. Five independent random seeds $\{42, 123, 456, 789, 2024\}$ are evaluated across all runs, with both partition masks and initializations redrawn for each. The architectural hyperparameters ($D = 64$, $H = 4$, dropout, learning rate, and schedule) are consistent with values conventional for HGT [72]. The loss coefficients in Equation (6) are specific to this task and were determined by informed judgment rather than tuning, as no convention exists for a five-term multi-task loss. We did not tune either the architectural or loss hyperparameters against the in-distribution test split or the LOSO folds; we did not search over them. The real-world evaluation in Section 7.4.1 is documented separately, as it operates at a different depth and epoch budget from other learned results in this paper. This method avoids selection leakage, but does not guarantee that either configuration is reported near its own optimum; the comparison is between untuned configurations, as explicitly stated.

4.2. Multi-Task Prediction Heads and Dimension Masking

From the final node embeddings $h_v^{(L)}$, SaG utilizes specialized multi-task prediction heads:

- **Reliability Head:** $\hat{R}(v) = \sigma(\text{MLP}_R(h_v)) \in [0, 1]$
- **Maintainability Head:** $\hat{M}(v) = \sigma(\text{MLP}_M(h_v)) \in [0, 1]$
- **Composite Failure Impact Head:** $\hat{I}^*(v) = \sigma(\text{MLP}_C(h_v \parallel \hat{R}(v) \parallel \hat{M}(v))) \in [0, 1]$
- **Relationship Criticality Head:** $\hat{Q}(u, v) = \sigma(\text{TypedEdgeEncoder}_{\phi(e)}(h_u, h_v, e_{uv})) \in [0, 1]$. This head is part of the released architecture but is *disabled throughout the evaluation reported here*. Every harness that produces a number in this paper instantiates the model with edge prediction switched off, so $\hat{Q}(u, v)$ is neither trained nor scored, and the edge term is absent from the objective below. We describe it for completeness with the implementation and make no edge-level claim on its basis.

4.2.1. Dimension-Masked Loss Formulation

The combined optimization objective integrates regression accuracy, multi-task dimension learning, ranking fidelity, and pairwise ordering:

$$\mathcal{L} = \mathcal{L}_{\text{composite}} + 0.5 \cdot \mathcal{L}_{\text{dimension}} + 0.3 \cdot \mathcal{L}_{\text{rank}} + 0.1 \cdot \mathcal{L}_{\text{pairwise}} + \lambda_{\text{RM}} \cdot \mathcal{L}_{\text{consistency}} \quad (6)$$

where $I^*(v)$ is the simulated cascade impact defined by the primary oracle (Section 4.3), $\mathcal{L}_{\text{composite}} = \text{MSE}(\hat{I}^*(v), I^*(v))$, $\mathcal{L}_{\text{rank}}$ is the ListMLE listwise ranking loss [83] parameterized by temperature τ :

$$\mathcal{L}_{\text{rank}} = -\frac{1}{N} \sum_{i=1}^N \left(\frac{\hat{s}_{\pi_i}}{\tau} - \log \sum_{j=i}^N \exp \left(\frac{\hat{s}_{\pi_j}}{\tau} \right) \right) \quad (7)$$

where $\pi = (\pi_1, \dots, \pi_N)$ denotes the permutation of nodes sorted in descending order of ground-truth impact $I^*(v)$, and $\hat{s}_v = \hat{I}^*(v)$. At the baseline default $\tau = 1.0$, the formulation reduces to standard ListMLE; the temperature parameter $\tau < 1.0$ is a configurable hyperparameter that sharpens probability distributions over narrow prediction margins. Pairwise ordering fidelity is guided by margin-ranking loss $\mathcal{L}_{\text{pairwise}} = \frac{1}{|P|} \sum_{(u,v) \in P} \max(0, \gamma - (\hat{s}_u - \hat{s}_v))$ with margin $\gamma = 0.05$ over pairs $P = \{(u, v) \mid I^*(u) - I^*(v) > \gamma\}$, and $\mathcal{L}_{\text{consistency}} = \text{MSE}([\hat{R}(v), \hat{M}(v)]_{v \in \text{unlabeled}}, [R_{\text{RM}}(v), M_{\text{RM}}(v)]_{v \in \text{unlabeled}})$ regresses predicted heads toward the diagnostic pathway’s baseline on unlabeled nodes, where $R_{\text{RM}}(v)$ and $M_{\text{RM}}(v)$ denote the deterministic Reliability and Maintainability scores from the explanation layer (Section 5). Headline results use $\lambda_{\text{RM}} = 0$, guaranteeing that the predictive and elucidative pathways remain strictly independent.

The coefficients in Eq. (6) (0.5 dimension, 0.3 listwise rank, 0.1 pairwise margin) balance composite regression with relative node ordering. Permutation-level ListMLE ($\mathcal{L}_{\text{rank}}$) provides the primary gradient force for global rank monotonicity (ρ), while pairwise margin ($\mathcal{L}_{\text{pairwise}}$) penalizes small-margin inversions among adjacent components. Across all seeds, gradient norms remain well-conditioned, preventing individual objectives from overpowering optimization.

Dimension Masking and Head Roles: Because dynamic cascade simulation ($I^*(v)$ via discrete-event cascade fault injection) observes runtime failure reachability rather than source-code maintainability, maintainability ground truth is unobserved during dynamic simulation. A separate change-propagation oracle $I_M(v)$ evaluates static structural change ripple at the Validate stage, but is never used as a training label to avoid circular supervision. We introduce a boolean dimension mask $m = [m_R, m_M] = [1, 0]$:

$$\mathcal{L}_{\text{dimension}} = \frac{1}{\sum_d m_d} \sum_{d \in \{R, M\}} m_d \cdot \text{MSE}(\hat{d}(v), d^*(v)) \quad (8)$$

This mask ensures the unobserved maintainability head is not artificially penalized or driven toward zero during backpropagation.

Auxiliary Nature of the Reliability Head and Active Loss: Under the headline experimental protocol, this general formulation simplifies significantly. With $\lambda_{\text{RM}} = 0$ (preserving strict pathway independence) and dimension mask $m = [1, 0]$, the cascade fault injection oracle emits a single scalar assigned to both targets: $R^*(v) = I^*(v)$ identically. $\mathcal{L}_{\text{dimension}}$ regresses $\hat{R}$ toward the same target as $\mathcal{L}_{\text{composite}}$. The effectively optimized loss reduces to:

$$\mathcal{L}_{\text{active}} = \text{MSE}(\hat{I}^*(v), I^*(v)) + 0.5 \cdot \text{MSE}(\hat{R}(v), I^*(v)) + 0.3 \cdot \mathcal{L}_{\text{rank}} + 0.1 \cdot \mathcal{L}_{\text{pairwise}} \quad (9)$$

Here, $\hat{R}$ re-enters the composite head as an input ($\hat{I}^* = \sigma(\text{MLP}_C(h_v \parallel \hat{R} \parallel \hat{M}))$), functioning as a feature-enrichment auxiliary pathway rather than independent multi-task supervision. We report the objective as implemented rather than claiming multi-dimensional supervisory ground truth.

4.2.2. Domain-Rewighted Criticality

ISO/IEC 25019’s Context of Use specifies that the relative weighting of reliability and maintainability is determined by deployment requirements rather than being fixed. The framework delivers a reweighting $\hat{Q}_{\text{domain}}(v) = q_R \hat{R}(v) + q_M M_{\text{static}}(v)$ to capture this flexibility. Because maintainability is unobserved during dynamic simulation ($m = [1, 0]$), headline results report $\hat{I}^*(v)$ directly; sensitivity relative to the static RM baseline is evaluated in Section 7.3.

4.3. Ground-Truth Simulation Oracles

To evaluate predictive accuracy before deployment without relying on production runtime telemetry, SaG executes discrete-event failure simulations over the raw structural multigraph $G_{\text{structural}}$. Table 3 summarizes the four component-level oracles and their distinct roles across the evaluation program.

- **Cascade Reachability Oracle ($I^*(v)$),** evaluated via discrete-event cascade fault injection: crashes component v , propagates outages across dependent topics, brokers, and links by breadth-first traversal, and returns the mean fractional feed loss over the subscriber population, each subscriber contributing

Table 3: Simulation oracles, operational constructs, and evaluation roles.

Oracle	Physical Mechanism	Nature	Role in Evaluation
$I^*(v)$	BFS cascade reachability + QoS ladder	Deterministic	Primary ranking target (RQ1, RQ2, RQ4)
$I_{\text{comp}}(v)$	Severity mixture: reachability + fragmentation	Deterministic	Explanation layer / Validate gate
$I_{\text{dyn}}(v)$	Discrete-event SimPy message queuing	Stochastic	Convergent-validity probe (RQ3)
$I_M(v)$	Reverse DEPENDS_ON traversal	Deterministic	Unsupervised maintainability reference

the unweighted mean loss of the topics it subscribes to. A topic’s feed loss is the fraction of its publishers that have failed (for a topic with no publisher, the fraction of its failed routers), scaled by a QoS ladder ($\times 1.2$ RELIABLE, $\times 1.15$ high/urgent priority, $\times 1.05$ medium) and clamped to $[0, 1]$. The denominator is the full subscriber set of the intact graph, so subscribers that themselves fail are retained in the average rather than excluded. The implementation admits a per-publisher rate weighting, but rates are declared per topic throughout our corpus, so it reduces exactly to this publisher fraction. This is the **primary continuous target label** throughout.

How much QoS is in this label. The ladder reads reliability and transport priority only; durability does not enter I^* at all, despite carrying the largest of the three QoS sub-weights in the framework’s own elicited vector (0.62, against 0.24 for reliability as well as 0.14 for priority; Section 3.2). That omission does not limit the label’s QoS content. Re-running the labeler with QoS scaling disabled entirely leaves the Application ordering very nearly intact — mean Spearman $\rho = 0.965$ against the ladder across the twelve folds (range 0.891–0.999) — and substituting a durability-aware $w(t)$ scaling moves it less still ($\rho = 0.977$). Neither parameterization materially reorders the target. The top- K critical set is the more sensitive construct: ladder and topology-only labels agree at mean Jaccard 0.678, so QoS does change *which* components are named critical without changing their order. I^* should therefore be read as a near-topological target that carries a QoS term at its threshold boundaries rather than through its ranking, which bounds what any QoS-encoding result can be crediting (Section 7.3.1).

- **Multi-Metric Composite Oracle** ($I_{\text{comp}}(v)$), evaluated via multi-metric failure simulation: a severity-weighted mixture of reachability loss, fragmentation, throughput loss and flow disruption, with AHP-derived coefficients (0.35, 0.25, 0.25, 0.15). Those coefficients come from a rank-one comparison matrix, so they record their origin without independently justifying them. They are not swept in our sensitivity analysis — a gap worth naming because I_{comp} supplies the labels for the explanation layer’s evaluation. It is reserved for Validate-stage gates and prescriptive verification, never for forecasting ranking.
- **Dynamic Queue-Flow Oracle** ($I_{\text{dyn}}(v)$), evaluated via discrete-event message-flow queue simulation (built on SimPy [84]): simulates emission rates, stochastic latencies, broker buffer saturation, and queue drops under fault injection, extracting the drop in delivered message rate to surviving consumers. To provide complete observability, the engine instruments Google Site Reliability Engineering (SRE)’s *Four Golden Signals* (latency, traffic, errors, saturation) across pre- and post-fault execution windows:
 - Latency*: decomposes end-to-end traversal latency (t_{e2e}) into private queue waiting time ($t_{\text{wait}} = t_{\text{dequeue}} - t_{\text{created}}$) and compute service time ($t_{\text{service}} = t_{\text{delivery}} - t_{\text{dequeue}}$), profiling p50, p95, and p99 percentiles;
 - Traffic*: monitors topic emission and subscriber delivery frequencies (Hz) alongside byte throughput (KB/s, Kbps);
 - Errors*: accounts for QoS deadline violations, queue overflow discards, best-effort network drops, and unserved message demand;
 - Saturation*: computes exact time-weighted mean queue depth ($\frac{1}{T} \int_0^T q_i(t) dt$), buffer occupancy ratios, and system-wide CPU utilization ($\rho_{\text{util}} \in [0, 1]$).

Crucially, golden signals are exposed as first-class diagnostic telemetry rather than mixed additively with $I_{\text{dyn}}(v)$. Under empirical testing, crashing a high-rate publisher clears downstream subscriber

queues (*contention relief*, $\rho = -0.499$ between delivery loss and tail latency delta). An additive composite would mathematically cancel delivery damage with latency reduction. $I_{\text{dyn}}(v)$ is therefore kept strictly 1-dimensional, serving as an independent convergent-validity probe (Section 7.3.2).

- **Change-Propagation Oracle** ($I_M(v)$), evaluated via structural change-propagation analysis: a deterministic reverse-dependency traversal over the transpose of the six-rule `DEPENDS_ON` projection, blending change reach, weighted change impact, and normalized depth. It is a structural maintainability reference and is never used as a training label, which would make the supervision circular.

Topic Criticality Label Masking. The multi-metric failure simulator can incorporate declared topic criticality into its severity term. This feature is disabled, however, because topic criticality is a GNN input feature: using it would result in the predictor being measured against a transformation of its own input.

Primary Oracle Declaration and Role Assignment. Because the three reliability-facing oracles (I^* , I_{comp} , I_{dyn}) measure separate operational constructs, we designate $I^*(v)$ (**cascade reachability injection**) as the **primary continuous target oracle** for all predictive ranking results (Tables 7–8, RQ1–RQ3). We select $I^*(v)$ over $I_{\text{dyn}}(v)$ for two methodological reasons. First, deterministic cascade reachability isolates structural dependency propagation with zero seed-to-seed variance, providing the reproducible ground truth required for deterministic CI/CD regression gating. Second, discrete-event queue simulation (I_{dyn}) introduces stochastic message latencies, bursty arrival distributions, and synthetic buffer limits that introduce queuing noise and workload assumptions, obscuring intrinsic architectural topology. $I_{\text{comp}}(v)$ is reserved for Validate-stage quality gates and prescriptive remediation verification, $I_{\text{dyn}}(v)$ serves as an independent convergent-validity probe (Section 7.3.2), and $I_M(v)$ serves as a structural maintainability reference.

Cross-Oracle Convergent Validity. As detailed in Section 7.3.2, the three reliability oracles show substantial but sub-ceiling agreement on Applications ($\rho = 0.627$ for (I_{dyn}, I^*) against a 0.811–1.000 label noise floor), confirming distinct constructs. Consequently, results established against one oracle are never transferred to another; every evaluation explicitly references its underlying simulation oracle.

4.4. Input–Label Independence Guarantee

To prevent data leakage, SaG applies strict architectural separation. Feature Space is constructed exclusively from G_{analysis} using static structural topology, static code metrics, and declared QoS contracts. Label Space is evaluated exclusively on raw $G_{\text{structural}}$ through independent simulation oracles (cascade reachability injection, composite failure simulation, and dynamic message-flow simulation). No simulation outputs, failure trace histories, or dynamic execution telemetry are exposed as input attributes to the GNN or the explanation layer.

What this guarantee does and does not establish

The separation rules out circular feature construction: no predictor can read a transformation of the quantity it is scored against. It does not establish probabilistic independence between features and labels, and we do not claim it does. G_{analysis} is a deterministic projection of $G_{\text{structural}}$ (Section 3.2). Hence, the labels are — up to the simulator seed — a deterministic function of the same topology from which the features are computed. Two consequences follow, and both bound the results of Section 7.

First, $I^*(v)$ is a topological functional, defined as a breadth-first reachability computation over $G_{\text{structural}}$ scaled by a QoS ladder. The predictive task is therefore to recover a closed-form graph function from features derived from the same graph. As a result, an unparameterized centrality score is expected to perform competitively with a trained model (Section 7.1). The observed parity is an anticipated outcome of the experimental design rather than an unexpected limitation of graph learning. We state this explicitly to guarantee clarity.

Second, and more restrictively, no result in this paper is validated against an observed failure. Every label — on synthetic topologies and on the five open-source systems alike — is simulator-derived. The evaluation can establish whether a learned model recovers a simulator’s ordering on architectures it was not trained on. Whether that ordering corresponds to which components actually fail in production is a question this design cannot answer, and Section 8.3 treats it as the study’s principal construct-validity threat.

5. The Explanation Layer: Standards-Grounded Criticality Attribution

The predictor described in Section 4 identifies risk concentration but does not address remediation strategies. SaG’s core claim is that a component may be critical because it is an unreplicated single point of failure, an error-propagating cascade hub, or a high-coupling maintainability bottleneck. Each structural cause requires a distinct remediation approach: broker replication, circuit-breaker insertion, or refactoring module dependencies. This section formalizes the diagnostic layer that attributes these causes. SaG decomposes component criticality into a standards-grounded quality profile, calculated using the same typed node properties (Section 3.4) but without parameter sharing with the neural predictor. It applies this profile to flagged components through triage rather than data flow (Figure 1).

This layer is explicitly unvalidated and intended as a design pattern for qualitative attribution rather than quantitative ranking. As shown in Section 7.1, its standalone rank correlation is low ($\rho = 0.205$, consistently below unweighted centrality). Its elicited AHP weights underperform a uniform prior (Section 7.3), and no human-subject studies have yet assessed developer adoption. This layer maps topological properties to standardized ISO/IEC concepts.

5.1. Grounding in ISO/IEC Standards

In accordance with ISO/IEC 25010:2023 [13] and ISO/IEC 25019:2023 [14], SaG formalizes two primary criticality dimensions. Component Criticality (D_1) is service loss upon component failure; Relationship Criticality (D_2) is service decline upon channel severance.

Criticality is assessed across two orthogonal characteristics: **Reliability** (R) and **Maintainability** (M). Reliability is divided into **Fault Tolerance** (FT) and **Availability** (A). FT uses Reverse PageRank, in-degree, and cascade depth potential to inform redundancy and circuit breaker strategies. A uses directed articulation points, bridge ratios, and connectivity degradation to inform replication strategies. Maintainability (M) assesses structural coupling and code-level complexity, using betweenness, QoS-weighted fan-out, code quality penalties, and clustering to guide decoupling and refactoring. This partition maps each ISO/IEC sub-characteristic to its graph metrics and remediation roles. Safety and security considerations that require specialized hazard logs are excluded from purely structural topology analysis.

5.2. Composite Quality Score Formulation

All raw metrics are rank-normalized to the interval $[0, 1]$ within the graph. Quality sub-characteristics are formulated hierarchically using the Analytic Hierarchy Process (AHP) [63]:

1. **Fault Tolerance** ($FT(v)$): Evaluates error cascade potential on transpose graph $G_{\text{analysis}}^\top$:

$$FT(v) = 0.45 \cdot \text{RPR}(v) + 0.30 \cdot \text{Deg}_{\text{in}}(v) + 0.25 \cdot \text{CDPot}_{\text{enh}}(v) \quad (10)$$

where $\text{RPR}(v)$ is Reverse PageRank (RPR), $\text{Deg}_{\text{in}}(v) = d_{\text{in}}(v)/(|V| - 1)$ is normalized in-degree, and $\text{CDPot}_{\text{enh}}(v) = \text{depth}(v)/\max_{u \in V} \text{depth}(u)$ is the normalized cascade depth potential, measuring the longest reachable directed failure-propagation chain from v on $G_{\text{analysis}}^\top$.

2. **Availability** ($A(v)$): Identifies structural single points of failure across five terms:

$$A(v) = 0.25 \cdot \text{AP}_c^{\text{dir}}(v) + 0.20 \cdot \text{QSPOF}(v) + 0.20 \cdot \text{BR}(v) + 0.25 \cdot \text{CDI}(v) + 0.10 \cdot w(v) \quad (11)$$

where $\text{AP}_c^{\text{dir}}(v)$ is Directed Articulation Point (AP) severity, $\text{QSPOF}(v)$ is QoS-weighted Single Point of Failure (QSPOF) severity, $\text{BR}(v)$ is Bridge Ratio (BR), $\text{CDI}(v)$ is Connectivity Degradation Index (CDI), and $w(v)$ is the intrinsic QoS weight.

3. **Reliability** ($R(v)$): Blends Fault Tolerance and Availability:

$$R(v) = r_{\text{FT}} \cdot FT(v) + (1 - r_{\text{FT}}) \cdot A(v), \quad r_{\text{FT}} = 0.36 \quad (12)$$

The intra-dimension weights apply $\lambda = 0.70$ shrinkage blending with a uniform prior. Because comparison matrices are rank-one by construction, these weights are documented conventions rather

than independently elicited consensus. The elicited AHP weights rank worse than a uniform prior against dynamic simulation (Section 7.3); whether they attribute better is untested, and we recommend the uniform prior pending a formal user study.

4. Maintainability ($M(v)$): Blends structural coupling with static code analysis:

$$M(v) = 0.35 \cdot \text{BT}(v) + 0.30 \cdot w_{\text{out}}(v) + 0.15 \cdot \text{CQP}(v) + 0.12 \cdot \text{CouplingRisk}_{\text{enh}}(v) + 0.08 \cdot (1 - \text{CC}(v)) \quad (13)$$

where $\text{BT}(v)$ is Betweenness Centrality (BT), $w_{\text{out}}(v)$ is QoS-weighted efferent coupling, $\text{CQP}(v)$ is Code Quality Penalty (CQP), and $\text{CC}(v)$ is local Clustering Coefficient (CC).

The baseline composite quality score integrates both dimensions as follows: $Q(v) = 0.80 \cdot R(v) + 0.20 \cdot M(v)$. When evaluated under an ISO/IEC 25019 Context of Use vector $\vec{\omega} = [q_R, q_M]^\top$, the score is dynamically reweighted: $Q_{\text{domain}}(v) = q_R \cdot R(v) + q_M \cdot M_{\text{static}}(v)$. Components are partitioned into Tukey tiers: CRITICAL ($Q > Q_3 + 1.5 \cdot \text{IQR}$), HIGH, MEDIUM, and MINIMAL. Across the benchmark topologies, this conservative Tukey upper fence flags an empirical mean of 4.2% of components (range 1.8%–8.3%), deliberately isolating the extreme right tail of architectural risk to prioritize developer intervention. High Availability (A) combined with low Fault Tolerance (FT) indicates a single point of failure that necessitates replication. In contrast, high Fault Tolerance (FT) identifies an error-cascade hub that requires circuit breakers (Section 8.4). Table 4 illustrates a representative Diagnostic Remediation Card generated for a flagged component, demonstrating how standardized quality sub-characteristics translate raw topological metrics into concrete refactoring actions.

Table 4: Illustrative Diagnostic Remediation Card for a flagged component (**TransactionProcessor** in the ATM System case study). The explanation layer maps structural metrics to ISO/IEC sub-characteristics to guide engineering intervention.

Diagnostic Attribute	Evaluated Value	Structural Interpretation
Component ID	TransactionProcessor	Core transaction coordinator (V_{app})
Tukey Risk Tier	CRITICAL	$Q(v) = 0.84 > Q_3 + 1.5 \cdot \text{IQR}$
<i>ISO/IEC 25010 Structural Quality Profile Breakdown</i>		
Availability (A)	0.89 [High SPOF Risk]	$\text{AP}_c^{\text{dir}} = 1.00$, $\text{QSPOF} = 0.92$, $\text{CDI} = 0.78$
Fault Tolerance (FT)	0.31 [Low Cascade Propagation]	$\text{RPR} = 0.25$, $\text{Deg}_{\text{in}} = 0.30$, $\text{CDPot} = 0.38$
Maintainability (M)	0.65 [Moderate Structural Coupling]	$\text{BT} = 0.70$, $w_{\text{out}} = 0.60$, $\text{CQP} = 0.40$
Structural Diagnosis	Isolated Single Point of Failure (Severe A deficit, low downstream cascade reach)	
Prescriptive Action	Deploy warm-standby replica and establish redundant message broker routing	

5.3. Prescriptive Remediation and Counterfactual Verification

After attributing root causes, candidate repairs (broker replication, circuit breaker insertion, topic decoupling) are generated and counterfactually verified in memory. Mutations are accepted only if they reduce systemic impact beyond simulation seed noise ($\Delta \bar{I}_{\text{comp}} > \kappa \cdot \sigma_{\text{seed}}$, $\kappa \geq 1.0$) without introducing new articulation points. This counterfactual verification loop illustrates the architectural pattern linking diagnosis to remediation; formal developer studies and automated patch synthesis benchmarks are reserved for future work.

6. Experimental Setup

6.1. Datasets and System Corpus

The evaluation corpus comprises 2,812 components across seventeen system architectures, including twelve synthetic topologies that form the inductive cross-validation folds and five real-world reference systems withheld from all training procedures, as detailed in Table 5. The synthetic scenarios span diverse operational domains (autonomous vehicles, financial trading, healthcare integration, industrial SCADA, smart-city IoT, telecom RAN, cloud microservices, and enterprise application integration via centralized broker hubs/ESB; detailed in Table S11 of the Supplementary Material).

Table 5: Overview of the evaluation corpus. The twelve synthetic topologies correspond to the inductive Leave-One-Scenario-Out folds described in Table 8, and the five real-world systems are excluded from all training folds and used exclusively for zero-shot transfer (Section 7.4). Per-scenario entity and edge counts are obtained from the committed topology files and verified through continuous integration.

Dataset	$ V $	$ V_{\text{app}} $	Topics	Brokers	Hosts	Libs	$ E $
Synthetic evaluation scenarios (11)	2,387	1,295	588	60	194	250	10,657
Synthetic case study — ATM (1)	74	26	27	5	8	8	261
Synthetic subtotal (12 LOSO folds)	2,461	1,321	615	65	202	258	10,918
Real-world subtotal (5 systems)	351	141	120	16	32	42	700
Total	2,812	1,462	735	81	234	300	11,618

6.1.1. Reproducibility of the Corpus

The benchmark corpus is designed to be fully regenerable rather than statically archived. Each dataset is deterministically generated from its configuration file using the following procedure:

```
python cli/generate_graph.py batch -input-dir data/scenarios -output-dir <path>
```

A companion manifest records the random seed, entity counts, git commit hash, and a SHA-256 cryptographic digest for each emitted topology. Continuous integration regression tests verify that every committed dataset regenerates byte-identically from its configuration and that all disk digests match the manifest. This procedure makes sure that third parties can reproduce the exact graphs used in these experiments, rather than sampling from similar distributions.

6.2. Baselines and Evaluated Predictors

Four primary predictor configurations, drawn from three distinct families, are evaluated alongside reference baselines. Table 6 provides a structured taxonomy of the evaluated predictors, their underlying graph representations, edge feature encodings, and empirical roles. Predictor names indicate the model family and substrate: an **-N** infix denotes a model trained on the complete native multigraph, its absence denotes the obtained Application–Library flow projection, and a **-QoS** suffix denotes a configuration that consumes declared QoS contracts. SaG throughout denotes the overall framework, not an individual predictor.

Table 6: Taxonomy of evaluated predictors and reference baselines. The **-N** infix denotes execution on the complete native multigraph; **-QoS** indicates inclusion of middleware QoS attributes.

Predictor	Evaluation Substrate	Typing	Edge Features	Parameters	Trained?	Empirical Role
Topo	G_{analysis} (Flow Projection)	No	None	0	No	Unweighted topological baseline
Topo-QoS	G_{analysis} (Flow Projection)	No	Scalar $w(e)$	0	No	QoS-weighted closed-form benchmark
GAT	G_{analysis} (Flow Projection)	Homogeneous	None	28,168	Yes	In-distribution counterpart of GAT-N (Table 7)
GAT-QoS	G_{analysis} (Flow Projection)	Homogeneous	Scalar $w(e)$	28,168	Yes	In-distribution counterpart of GAT-N-QoS (Table 7)
GAT-N	Native Multigraph	Homogeneous	None	28,168	Yes	Untyped, unweighted GNN floor
GAT-N-QoS	Native Multigraph	Homogeneous	Scalar $w(e)$	28,168	Yes	Isolates QoS channel without typing
HGT	Native Multigraph	Heterogeneous	Relation 1-hot	434,620	Yes	Isolates relational typing without QoS
HGT-QoS	Native Multigraph	Heterogeneous	16-D QoS Vector	434,620	Yes	Proposed full learned model
RM / $Q(v)$	G_{analysis} (Analysis Graph)	Per-type rules	Scalar $w(e)$	0	No	Diagnostic attribution reference

Table 6 summarizes the evaluated predictors, graph representations, and empirical roles across three families: heterogeneous graph learning (**HGT-QoS** with 16-D QoS edge vectors, and its ablation **HGT**), homogeneous graph learning (**GAT-N-QoS** with scalar $w(e)$, and unweighted **GAT-N**), and training-free structural baselines (**Topo-QoS** and unweighted **Topo** on the **DEPENDS_ON** flow projection). The **-N** infix denotes the native multigraph substrate; on flow projections, homogeneous variants are denoted **GAT/GAT-QoS** (Table 7). The out-of-distribution evaluation (Table 8) additionally reports **RM** ($Q(v)$, Section 5) as a diagnostic reference baseline. Deterministic RM scoring also drives sensitivity sweeps in Section 7.3.

6.2.1. Evaluation Substrates and Parity Guarantee

Predictors in this study operate on matched substrates within their respective families:

Graph Learning Models (GAT-N-QoS, HGT-QoS): Under Leave-One-Scenario-Out (Table 8), both learned predictors ingest the complete native typed multigraph across all five entity types (recorded using the

shared -N infix). In-distribution (Table 7), GAT/GAT-QoS consume the Application–Library `DEPENDS_ON` projection, confounding typing with multi-entity visibility. GAT-N-QoS uses per-type projection with untyped GATConv across edges, whereas HGT-QoS uses relation-specific HGTConv weights. The edge channel also differs: GAT-N-QoS consumes a scalar $w(e)$, while HGT-QoS consumes all 16 dimensions. The parameter budget (434,620 vs. 28,168) and directionality also remain open confounds (Section 8.4).

Training-Free Structural Baselines (Topo, Topo-QoS): We evaluate topological baselines on the obtained Application–Library `DEPENDS_ON` projection (Section 3.2), since raw multigraphs route messages via topics/brokers, leaving Application nodes with near-zero betweenness.

Ground-Truth Independence Guarantee: Crucially, **no predictor in either group accesses $G_{\text{structural}}$** : that raw topology is strictly reserved for simulation oracles (Section 4.4), verified by `tests/test_independence_guarantee.py`. Regardless of substrate, all variants are scored on the same independently resolved Application node set (V_{app} , Section 6.3).

6.3. Evaluation Metrics and Protocols

Figure accessibility. All figures employ the Okabe–Ito palette with distinct markers and hatchings to ensure monochrome legibility.

Ranking Precision: Evaluated via Spearman ρ and Kendall τ against the simulated impact $I^*(v)$ from the primary oracle (Section 4.3). **Critical-Set Identification:** Measured via $F_1@K$, Precision@ K , and Recall@ K for top- K components ($K = \text{round}(0.20 \cdot |V_{\text{app}}|)$). Because the predicted and reference sets are both of size K , the three coincide identically and reduce to top- K set overlap. **Statistical Significance:** Paired Wilcoxon signed-rank tests [85] ($p < 0.05$) and bootstrap 95% CIs ($B = 2,000$) over folds [86, 87]. In the 12-fold LOSO design, the power floor is $p = 0.00049$. Applying Holm’s step-down correction across ten full-population rank contrasts (Sections 7.1–7.3.1), three survive: **Topo-QoS** over **Topo** ($p = 0.0010$), **Topo** over **RM** ($p = 0.0005$), and unweighted typing **HGT** over **GAT-N** ($p = 0.0005$). The two contrasts that carry the QoS channel into an already-typed model do not: typing with the QoS channel present reaches only $p = 0.1294$ (9/12 folds) and the QoS edge ablation under typing $p = 0.2036$ (10/12), as Table 10 reports and Section 7.3.1 discusses.

Pre-registration. The primary out-of-distribution contrast (HGT-QoS vs. Topo-QoS under LOSO, 5 fixed seeds, fold as unit of analysis) was pre-registered in the replication package prior to obtaining results. As reported in Section 7.1, the margin did not reach statistical significance.

6.3.1. Evaluation Population and Protocols

Each predictor within an evaluation table is scored on an identical node population, resolved strictly from scenario topology and ground truth, specifically the **Application** set (V_{app}) unless otherwise noted. Pooling node types conflates distinct base rates and can trigger Simpson’s paradox (Section 7.3).

In-Distribution Evaluation: Stratified 60% train / 20% val / 20% test node splits over five seeds {42, 123, 456, 789, 2024}, redrawing partitions and initializations. **Inductive LOSO Cross-Validation:** Models are trained on eleven scenarios and test zero-shot on the held-out twelfth across all 12 folds under equal 3-layer depth and inner-split early stopping (Section 8.4). **Real-World Architectural Transfer:** Synthetic-trained models are evaluated zero-shot on five open-source systems without fine-tuning.

7. Results and Empirical Analysis

Empirical results for RQ1–RQ5 are presented across the twelve-fold inductive benchmark and five authentic open-source distributed systems. The evaluated populations are strictly stratified on the Application service set (V_{app}) under the input–label independence guarantee (Section 4.4).

7.1. RQ1: Graph Learning vs. Structural Baselines

RQ1 Summary (Predictive Efficacy): **HGT-QoS** ($\rho = 0.638$) achieves highest numerical correlation but does not significantly outperform the unparameterized **Topo-QoS** baseline ($\rho = 0.553$, $\Delta\rho = +0.085$, $p = 0.151$, 95% CI spanning zero). Because ground-truth reachability $I^*(v)$ is a topological functional, closed-form

QoS-weighted centrality captures broad failure propagation without training. Restricting evaluation to active components halves correlation across all models ($\rho_{>0} \approx 0.36$), showing that full-population correlation largely reflects separation of inert sink nodes.

Table 7 presents in-distribution held-out performance versus simulated cascade impact $I^*(v)$ across all twelve distributed architecture scenarios ($n = 12$), evaluating both global rank-order monotonicity (Spearman ρ) and operational critical-set identification ($F_1@K$ on the top 20% most vulnerable Application services).

Table 7: In-distribution held-out evaluation across all twelve distributed architecture scenarios, reporting both Spearman rank correlation (ρ) and critical-set identification ($F_1@K$, with $K = \text{round}(0.20 \cdot n)$): mean over five seeds; n = held-out Application test count. Because both the predicted and the reference set have exactly K members, precision, recall and $F_1@K$ coincide identically, so this column measures top- K overlap rather than identification against an independently sized critical set (Section 6.3). Substrates differ in-distribution and are listed per predictor in Table 6 (Section 6.2). Each seed redraws the 60/20/20 split and model initialization. The held-out set is small on several scenarios — $n = 5$ on ATM, giving $K = 1$ — so a single rank swap moves ρ substantially and the per-scenario cells on the smaller topologies are coarsely quantized; they are reported for completeness and the row means, not individual cells, carry the comparison.

Scenario	n	Topo		Topo-QoS		GAT		GAT-QoS		HGT		HGT-QoS	
		ρ	F_1	ρ	F_1	ρ	F_1	ρ	F_1	ρ	F_1	ρ	F_1
ATM System	5	0.538	0.400	0.557	0.400	-0.393	0.000	-0.080	0.000	0.492	0.600	0.348	0.400
AV System	16	0.188	0.333	0.797	0.533	0.816	0.400	0.465	0.267	0.637	0.533	0.558	0.533
Enterprise	60	0.443	0.600	0.793	0.600	0.779	0.583	0.481	0.433	0.861	0.600	0.878	0.600
Financial Trading	12	0.387	0.200	0.512	0.400	0.565	0.400	0.666	0.400	0.693	0.500	0.730	0.500
Healthcare	10	0.291	0.200	0.399	0.000	0.725	0.300	0.575	0.300	0.575	0.400	0.607	0.500
Enterprise Integration (ESB)	14	0.179	0.267	0.429	0.400	0.363	0.467	-0.156	0.067	0.421	0.400	0.476	0.400
Industrial SCADA	28	0.601	0.533	0.710	0.533	0.656	0.533	0.478	0.500	0.787	0.667	0.839	0.633
IoT Smart City	40	0.320	0.350	0.397	0.350	0.580	0.450	0.538	0.425	0.849	0.650	0.850	0.650
Logistics Fleet	22	0.511	0.500	0.652	0.400	0.746	0.500	0.780	0.550	0.796	0.550	0.815	0.500
Microservices	18	0.219	0.150	0.344	0.250	0.351	0.400	0.363	0.450	0.141	0.300	0.664	0.600
Real-Time Gaming	15	0.360	0.533	0.802	0.533	0.464	0.333	0.471	0.400	0.651	0.533	0.641	0.400
Telecom RAN	24	0.402	0.480	0.422	0.280	0.608	0.320	0.350	0.360	0.591	0.280	0.526	0.320
Mean	—	0.370	0.379	0.568	0.390	0.522	0.391	0.411	0.346	0.624	0.501	0.661	0.503

In-distribution critical-set identification provides diagnostic insight into model fitting across architectures. On training graphs, heterogeneous architectures achieve higher in-distribution fit (HGT mean $F_1 = 0.501$, HGT-QoS 0.503) than homogeneous message passing (GAT 0.391, GAT-QoS 0.346) and structural baselines (Topo 0.379, Topo-QoS 0.390), demonstrating higher representational capacity on seen graphs rather than inductive transfer. Notably, in-distribution F_1 evaluation reveals fitting behaviors obscured by rank correlation alone. In Healthcare, Topo-QoS achieves $\rho = 0.399$ but fails at critical triage ($F_1 = 0.000$); in Microservices, unaugmented HGT suffers degradation ($\rho = 0.141$, $F_1 = 0.300$), whereas HGT-QoS recovers both ranking monotonicity ($\rho = 0.664$) and top- K overlap ($F_1@K = 0.600$). These $F_1@K$ figures are set-overlap measures, not recall against an independently sized critical set: both sets have exactly K members, so precision, recall and F_1 take the same value by construction (Section 6.3).

7.1.1. Out-of-Distribution (LOSO) Generalization

In inductive Leave-One-Scenario-Out (LOSO) cross-validation, models are assessed based on their ability to predict cascading criticality across entirely unseen system topologies:

We evaluated twelve LOSO folds, encompassing eleven synthetic scenarios and the ATM case study, as detailed in Table 5 (Section 6.1). In each fold, we held out one scenario for zero-shot testing and trained the model exclusively on the remaining eleven. We assessed all model variants on the same Application node set per fold (Section 6.3) and ran paired Wilcoxon tests across the twelve folds, yielding a minimum attainable two-sided p -value of 0.00049. The evaluated populations per fold ranged from 26 to 300 Application nodes, resulting in $K = \text{round}(0.20 \cdot |V_{\text{app}}|)$ values between 5 and 60. For $F_1@K$, HGT-QoS achieved 0.424 compared to Topo-QoS’s 0.388 ($\Delta = +0.037$, prevailing in 7 of 12 folds, $W = 29.0$, $p = 0.470$). The untyped GAT-N-QoS attained 0.431 ($\Delta = -0.006$, prevailing in 5 of 12 folds, 1 tie, $W = 27.5$, $p = 0.653$), indicating that critical-set identification does not statistically distinguish the typed model from either untyped learning or the QoS-weighted baseline.

Label-noise ceiling. These correlations are bounded by the reproducibility of the target they are scored against. Re-running the ground-truth oracle across the five seeds gives a test-retest rank correlation between

Table 8: Inductive LOSO evaluation, Application population, twelve folds. Learned variants share training set, the native multigraph substrate (hence the **-N** infix; Section 6.2), depth, and selection rule (Section 6.3), differing in typing and edge channel; parameter budget and message-passing directionality remain unmatched confounds that Table 10 controls for (Section 8.4). Fold score = mean over five seeds; **Fold** σ = spread of the twelve fold means; **Seed** σ = median within-fold spread (zero by construction for deterministic baselines); CI = bootstrap over folds ($B = 2,000$). $\Delta\rho$ is paired by fold against **Topo-QoS**, the pre-registered comparator (Section 6.3), with its own bootstrap interval; an interval spanning zero means the contrast is not resolved at twelve folds, which is the case for every learned variant. **Critical-Set** $F_1@K$ is top- K set overlap: the predicted and reference sets are both of size K , so precision, recall and F_1 coincide identically (Section 6.3).

Predictor / Reference	Mean LOSO ρ	95% CI	$\Delta\rho$ vs Topo-QoS	Fold σ	Seed σ	Critical-Set $F_1@K$	Requires Training
<i>Training-free structural baselines</i>							
Topo	0.349	[0.254, 0.452]	-0.204 [-0.286, -0.122]	0.173	—	0.366	No
Topo-QoS	0.553	[0.443, 0.657]	— (reference)	0.192	—	0.388	No
<i>Learned predictors (shared native substrate, matched training set)</i>							
GAT-N	0.317	[0.254, 0.381]	-0.236 [-0.342, -0.125]	0.111	0.298	0.328	Yes
GAT-N-QoS	0.604	[0.538, 0.665]	+0.051 [-0.067, +0.169]	0.112	0.024	0.431	Yes
HGT	0.551	[0.474, 0.617]	-0.002 [-0.066, +0.072]	0.124	0.114	0.427	Yes
HGT-QoS	0.638	[0.561, 0.710]	+0.085 [-0.029, +0.194]	0.133	0.052	0.424	Yes
<i>Diagnostic reference — not a ranking model</i>							
RM / $Q(v)$	0.205	[0.092, 0.320]	-0.348 [-0.432, -0.265]	0.195	—	0.322	No

0.811 and 1.000 across the twelve folds (median 0.982; nine of twelve at or above 0.95), with Microservices the least reproducible at 0.811. **HGT-QoS's** $\rho = 0.638$ therefore recovers roughly 65% of the attainable signal against the median ceiling, and no predictor in Table 8 can exceed the reproducibility of its own labels. Top- K critical sets are a much noisier construct: their cross-seed Jaccard has a median of 0.847 and falls to 0.370 (Logistics Fleet), 0.500 (Industrial SCADA), and 0.500 (Telecom RAN). That instability is the main reason the $F_1@K$ margins are less stable than the ranking margins, and it bounds how much weight any single critical-set comparison can carry. Microservices is the least reproducible fold, but it is not one the typed model loses (Section 7.2.1): **HGT-QoS** wins it by +0.229. On this corpus, the folds with the lowest label ceiling and the folds where the model is beaten are disjoint.

Figure 3 summarizes these results alongside critical-set identification and inter-oracle agreement.

Key Insights concerning RQ1:

- Typed learning is the best configuration, but not demonstrably better than the QoS baseline.** **HGT-QoS** leads all predictors out-of-distribution ($\rho = 0.638$). Against training-free **Topo-QoS** it is +0.085 (9/12, $W = 20.0$, $p = 0.151$, CI [-0.029, +0.194]), an interval that includes zero; un-augmented **HGT** is indistinguishable from the baseline outright (-0.002, 3/12, $p = 0.470$). Neither pre-registered contrast reaches significance, and we report that as the answer rather than as a near miss.
- A QoS-weighted structural score is a genuinely strong baseline — and untyped learning is worse than it.** **Topo-QoS** reaches $\rho = 0.553$ zero-shot, beating unweighted **Topo** on all twelve folds (+0.204, $p = 0.0005$). More pointedly, the untyped, unweighted learned model loses to it decisively (**GAT-N**, -0.236, 2/12, $p = 0.0024$). On this task, a homogeneous graph network trained on eleven architectures does not reach what a closed-form centrality score achieves with no training at all. Any claim that graph learning is required must be made against this baseline.
- Critical-set identification does not favor the typed model.** On $F_1@K$, **HGT-QoS** scores 0.424 against **GAT-N-QoS's** 0.431 and **HGT's** 0.427 — a three-way tie within noise — while all three numerically lead **Topo-QoS** (0.388), though without statistical significance throughout folds ($\Delta = +0.037$, $p = 0.470$). The margin over untyped learning claimed in earlier versions does not hold on the reconciled 12-fold corpus.
- Power is not the limiting factor.** At $n = 12$, the design tolerates four lost folds and still reaches $\alpha = 0.05$, provided the losses are smallest in magnitude. **HGT-QoS's** are not: it loses Enterprise (-0.335) and Telecom RAN (-0.169) to **Topo-QoS** by the two largest margins in the set, which is what holds W at 20.0. Enlarging the corpus will not resolve this; we must understand the inversions instead (Section 7.2.1).
- The explanation layer is weakly predictive, not noise.** **RM/ $Q(v)$** reaches $\rho = 0.205$, losing to unweighted **Topo** on every fold (-0.144), so no ranking claim is made for it. Its interval [0.092, 0.320]

stays above zero, and it supplies interpretable diagnostics without training (Section 5). Table 8 lists it as a reference point, not a competitor.

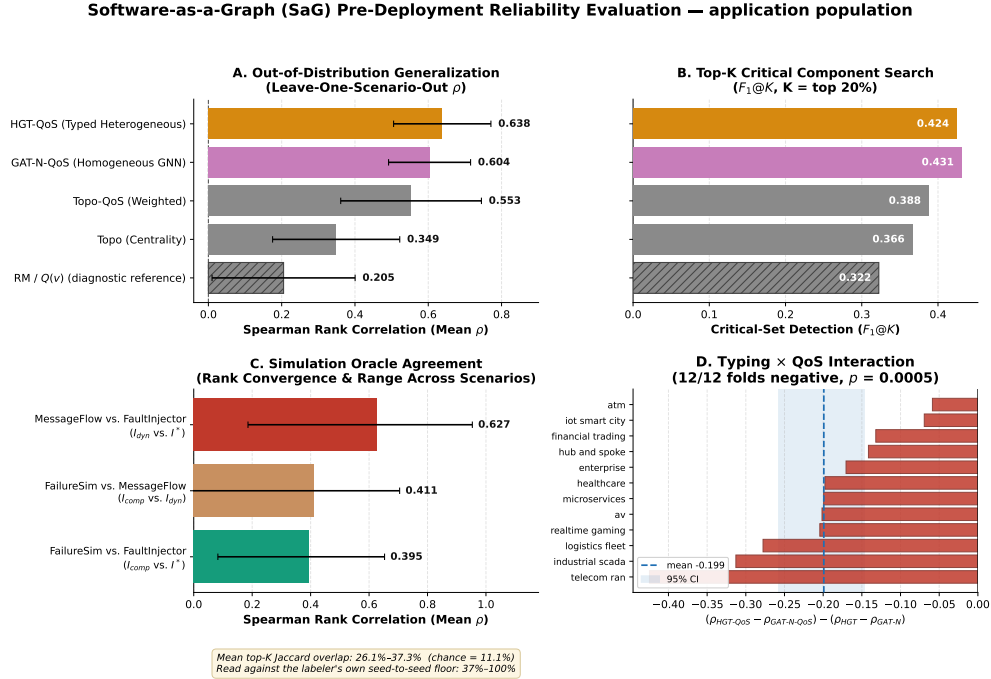


Figure 3: Results at a glance: Application population. (A) Out-of-distribution rank correlation per predictor across the twelve LOSO folds. (B) Critical-set identification at $K = 20\%$, where the typed model does not separate from untyped learning. (C) Pairwise rank agreement between the three simulation oracles, against the chance baseline. (D) The typing $\times$ QoS interaction per fold, negative on all twelve (Section 7.2); the dashed line is the mean and the band its bootstrap 95% CI. Panels A and B are read from the same artifact as Table 8, C from the convergent-validity artifact, and D from the significance artifact behind Table 10.

7.1.2. The Active Stratum: Ranking Quality Separated From Inertness Detection

Every LOSO figure above shows correlation over the full held-out Application population, and between 21% (Microservices) and 52% (Healthcare) of that population carries exactly zero simulated impact, depending on the fold. A predictor can therefore score well by separating components that can propagate a failure from those that cannot, without ordering the propagating ones correctly. Because these are different capabilities with different functional value, we re-score all twelve folds on the *active stratum* — the $n_{>0}$ components with strictly positive ground-truth impact — using the same predictions, folds, and seeds. Table 9 reports both.

Table 9: LOSO means over twelve folds on the full Application population (ρ) and restricted to components with strictly positive ground-truth impact ($\rho_{>0}$). Same predictions, folds, and seeds; only the evaluated subset differs. Retained is defined as the active-stratum ratio $\rho_{>0}/\rho$ (percentage of full-population correlation preserved).

Predictor	ρ (full)	$\rho_{>0}$ (active)	Retained
RM / $Q(v)$	0.205	0.102	49%
Topo	0.349	0.181	52%
Topo-QoS	0.553	0.280	51%
GAT-N	0.317	0.159	50%
GAT-N-QoS	0.604	0.328	54%
HGT	0.551	0.299	54%
HGT-QoS	0.638	0.356	56%

Two consequences follow, the first of which revises a claim made in earlier versions of this paper.

1. **The restriction costs every predictor roughly half its correlation, learned or not.** Retained fractions run from 49% (RM) to 56% (HGT-QoS) with no systematic separation between the training-free and learned families (Topo-QoS 51%, GAT-N 50%, HGT 54%). Roughly half of every predictor’s full-population correlation reflects separating inert components from active ones, a property of the label distribution rather than a discriminator between methods.
2. **The method ordering is unchanged, and so are the verdicts.** On the active stratum HGT-QoS still leads ($\rho_{>0} = 0.356$), ahead of Topo-QoS (0.280) and GAT-N-QoS (0.328). Nothing in Section 7.1 or Section 7.2 turns on whether zero-impact components are included; we report both columns because they answer alternative operational questions.

7.2. RQ2: Value of Typed Heterogeneity

RQ2 Summary (Typing vs. QoS):. Relational typing and continuous QoS edge encodings act as empirical substitutes rather than complements. While each mechanism individually exhibits a significant main effect over unweighted baselines (+0.134 and +0.187, Holm $p = 0.0015$), their factorial interaction is sub-additive ($\Delta\rho = -0.199$, negative on all 12 folds, $p = 0.0005$). In pub-sub architectures, QoS contracts align with communication relations; thus, edge features and relation-specific weights supply redundant traversal indicators, yielding diminishing returns when combined.

RQ2 investigates whether relation typing boosts prediction compared to homogeneous message passing. Addressing this question requires ablating typing against a matched comparator, and the outcome depends entirely on the comparator selected:

- **Both factors have a real main effect.** Averaged over the other factor’s levels, relation typing is worth $\Delta\rho = +0.134$ (12 of 12 folds, $W = 0.0$, $p = 0.0005$, Holm 0.0015) and the QoS edge channel +0.187 (11 of 12, $p = 0.0015$, Holm 0.0015), subject to the capacity and directionality confounds detailed below.
- **But they interact, strongly and sub-additively.** The difference of differences — how much typing buys with the QoS channel present, minus how much it buys without it — is -0.199 , negative on all twelve folds ($W = 0.0$, $p = 0.0005$, Holm 0.0015, 95% CI $[-0.258, -0.147]$).
- **The simple effects show the same pattern on both sides.** Typing is worth +0.234 when the QoS channel is absent (12/12, $p = 0.0005$) and +0.035 when it is present ($p = 0.1294$); the QoS channel is worth +0.287 without typing ($p = 0.0010$) and +0.087 with it ($p = 0.2036$).
- **In-distribution fitting (Table 7) is not evidence either way.** The homogeneous pair reads the Application–Library projection while the typed pair reads the native multigraph, confounding message passing with multi-entity visibility.

Typing and QoS encoding are substitutes, not complements. Each mechanism alone lifts the plain baseline from $\rho = 0.317$ to 0.55–0.60; together they reach 0.638, barely more than either achieves alone. Both supply identical information: which relation an edge traverses. Weight matrices encode it for typing; edge vectors encode it for QoS. Furthermore, $I^*(v)$ ’s ordering is recovered at mean $\rho = 0.965$ by topology-only relabeling without QoS terms (Section 4.3), confirming neither channel tracks impact unexpressed by the oracle.

Reference arm and confounders. Both large simple effects are measured against GAT-N, an unweighted floor ($\rho = 0.317$, seed $\sigma = 0.298$). While substrate, training set, depth, and early stopping are held constant, parameter capacity (434,620 in HGT-QoS vs. 28,168 in GAT-N-QoS, 15.4 $\times$) and message directionality (103,725 reverse-relation parameters in HGTConv) remain unmatched. The +0.134 typing margin reflects this joint architectural transition, requiring registered controls (GAT-N-C, GAT-N-QoS-C, HGT-QoS-U, Section 8.4) to isolate. On this evidence the 15.4 $\times$ capacity gap buys a non-significant margin.

7.2.1. Where Typed Learning Fails, and How It Can Be Detected

HGT-QoS loses to Topo-QoS on three folds, but only two of them are substantive: Enterprise ($\rho = 0.461$ vs. 0.795) and Telecom RAN (0.407 vs. 0.576). The third, AV System (0.722 vs. 0.753), is a near-tie at -0.030 and carries little interpretive weight. The two substantive inversions are what keep the RQ1 comparison

Table 10: The 2×2 over relation typing (T) and the QoS edge channel (Q), whose four cells are the four reported learned arms: GAT-N ($\neg T \neg Q$), HGT ($T \neg Q$), GAT-N-QoS ($\neg T Q$), HGT-QoS ($T Q$). **Holm correction is applied across the three orthogonal quantities in the upper block only**; the four simple effects below are algebraically determined by those three and are reported descriptively (Section 7.3.1). Main effects average over the other factor’s levels; the typing effect reflects the joint transition to HGT and is confounded (Section 8.4). **Won** counts folds with $\Delta > 0$; the interaction is *negative* on all twelve. All quantities are post-hoc, and none were pre-registered. Fold overlap makes the reported p -values nominal (Section 8.3).

Quantity	Contrast	$\Delta\rho$	Won	W	p	p_{Holm}
<i>The 2×2: three orthogonal quantities, Holm-corrected across these three</i>						
Typing (main effect)	averaged over Q	+0.134	12/12	0.0	0.0005	0.0015
QoS channel (main effect)	averaged over T	+0.187	11/12	2.0	0.0015	0.0015
Typing $\times$ QoS interaction	difference of differences	−0.199	0/12	0.0	0.0005	0.0015
<i>Simple effects — descriptive, not separately corrected</i>						
Typing, QoS absent	HGT vs. GAT-N	+0.234	12/12	0.0	0.0005	—
Typing, QoS present	HGT-QoS vs. GAT-N-QoS	+0.035	9/12	19.0	0.1294	—
QoS channel, typing absent	GAT-N-QoS vs. GAT-N	+0.287	11/12	1.0	0.0010	—
QoS channel, typing present	HGT-QoS vs. HGT	+0.087	10/12	22.0	0.2036	—

below significance. All three losses fall where **Topo-QoS** is at its own strongest. Enterprise, AV and Telecom RAN are its 2nd, 3rd and 7th best folds of twelve (0.795, 0.753, 0.576), each at or above its mean of 0.553. This supports reading the inversions as the learned model discarding structural signal the baseline retains, rather than as folds that are intrinsically hard — on the two folds where **Topo-QoS** is weakest (Microservices 0.265, ATM 0.311) the typed model wins by +0.229 and +0.210.

Absence of a Label-Free Confidence Signature. We evaluated whether the standard deviation of predicted scores $\hat{\sigma}$ over held-out applications may indicate model trustworthiness at inference time without labels. It does not. Low dispersion does not mark the folds the model loses, and the lowest-dispersion folds include ones carrying large positive margins, so no threshold on $\hat{\sigma}$ separates the two. We report the direction of this result rather than its coefficients. The dispersion diagnostic was computed against a superseded prediction export, and none exists at the fidelity of the twelve-fold artifact behind Tables 5–7, so the per-fold $\hat{\sigma}$ values are withheld pending a re-run. Graph size is measurable on the current artifact and also fails to flag difficulty in advance (rank correlation with margin -0.434 , $p = 0.159$), as does connection density.

Feature scale drift across scenarios remains the primary explanation for the Enterprise deficit, as Enterprise is the largest graph (520 nodes) and feature-scaling disparities are most acute in this case. Because the protocol strictly holds model hyperparameters constant across folds, there is currently no automated, label-free signal to predetermine whether an unseen architecture will favor the learned model or the training-free baseline (Section 8.4).

7.3. RQ3: Ablations and Sensitivity Analysis

RQ3 Summary (Ablations and Sensitivity): Only two of the explanation layer’s ten declared constants matter under Morris screening, and no configuration alters any comparison above. The elicited AHP weights are anti-predictive against a uniform prior, and the three reliability oracles agree substantially but below the label-noise ceiling.

This section presents ablations relevant to the primary claims, including the QoS edge encoding, cross-oracle agreement, and per-type stratification that informs the interpretation of the results that follow. Of the ten constants, only the AHP shrinkage λ and the Fault-Tolerance/Availability blend r_{FT} exhibit appreciable influence on ρ ($\mu^* = 0.134$ and 0.132 under Morris screening, compared to ≤ 0.025 for the remaining eight). No configuration of the topic-weight or QoS sub-weight constants alters any comparison reported above. The elicited AHP weights are, notably, *anti*-predictive: rank correlation decreases monotonically from 0.319 under a uniform setting to 0.200 under raw AHP judgment. We retain these weights because RM serves as an attribution instrument rather than a ranking model, a trade-off discussed in Section 8.4.

7.3.1. QoS Feature Ablation

To isolate the specific empirical contribution of the continuous-categorical QoS edge features (Section 4.1.1), we evaluated **HGT**, an unaugmented ablation of HGT-QoS whose edge features contain only scalar coupling and relation one-hot encodings.

Under the inductive LOSO evaluation, the QoS edge encoding’s value depends on whether relation typing is already present. This is not a second finding but the same one seen from the other side. The interaction tested in Section 7.2 (-0.199 , negative on all twelve folds, $p = 0.0005$) is symmetric in the two factors. Hence, a conditional effect of typing on the QoS channel is necessarily also a conditional effect of the QoS channel on typing. The figures below are the simple effects of Table 10 restated for the ablation reader; the significance of their *difference* rests on that interaction, not on the difference between their p -values.

Without typing, the encoding is decisive. GAT-N-QoS reaches $\rho = 0.604$ against GAT-N’s 0.317: $\Delta\rho = +0.287$, won in 11 of 12 folds, $W = 1.0$, $p = 0.0010$, CI $[+0.207, +0.365]$. *With typing, it is not significant.* HGT-QoS reaches 0.638 against HGT’s 0.551: $+0.087$, 10 of 12 folds, $W = 22.0$, $p = 0.2036$. An earlier version of this manuscript reported the typed gain as $+0.054$ at $p = 0.0093$ from a superseded artifact and treated it as an independent contribution on top of typing; the effect does not reproduce at that significance, and the independence claim is withdrawn.

The encodings also improve optimization reproducibility, and here the asymmetry runs the other way. The median within-fold standard deviation across five seeds is 0.024 for GAT-N-QoS against 0.298 for GAT-N — more than a tenfold reduction — and 0.052 for HGT-QoS against 0.114 for HGT. An untyped model without the QoS channel is the least stable configuration by a wide margin, and either mechanism stabilizes it. This is consistent with the ranking result: both channels tell the model which relation an edge belongs to, and a model given neither must infer it from topology alone.

The target is nearly QoS-free, which bounds what either channel can credit.. The gains above are earned against $I^*(v)$, whose ordering a topology-only relabeling recovers at mean $\rho = 0.965$ across the same twelve folds, with no QoS term in the labeler at all (Section 4.3). The QoS edge channel therefore cannot help the model track QoS-driven impact that the oracle does not itself express, which is the strongest evidence we have for reading it as a relation-identity channel rather than a contract-semantics one. The label does move under QoS, but only at its top- K boundary (Jaccard 0.678 against the topology-only arm), not in its ranking. A corpus whose oracle expressed QoS-driven impact in its ordering — deadline misses, durability replay, priority inversion under load, none of which I^* observes — would be a stronger test of the encodings than the one we report.

QoS Parameter Variance. Modal QoS shares range from 29% to 89% across the twelve scenarios, making sure that every fold carries genuine variation in declared reliability, durability, and priority. As noted in Section 4.1.1, one schema dimension (`max_blocking_ms_log`) remains zero throughout the corpus as a reserved extension point, while the declared deadline populates the other two (`has_deadline`, `deadline_ns_log`) on 75% of topics. Reported gains therefore stem from six active dimensions.

7.3.2. Convergent Validity Over Simulation Oracles

The three reliability-facing oracles measure distinct constructs, so we checked whether they agree before treating any as ground truth. Over the twelve inductive folds on the Application population, the behavioural queue-flow oracle and the topological cascade injector agree at mean Spearman $\rho = 0.627$ (top- K Jaccard 0.370 against 0.111 expected by chance), against I^* ’s own seed-to-seed test-retest of 0.811–1.000. The agreement is therefore substantial but distinctly below label noise, which is the reading we want: an oracle reproducing another to within its own reproducibility would be re-measuring the same topology rather than corroborating it. Two boundaries qualify this — a large share of the agreement is the two oracles concurring on which components are *harmless*, and I_{dyn} has a measured noise floor of its own that the headline does not correct for.

Simultaneously, the Four Golden Signals captured during discrete-event execution reveal the physical mechanism underlying this construct separation. When a critical publisher crashes, surviving consumers experience substantial delivery degradation ($I_{\text{dyn}} > 0$), yet downstream queue wait times and buffer

saturation drop markedly due to contention relief ($\rho = -0.499$ with tail latency delta). Importantly, $I^*(v)$ is defined strictly through deterministic topological reachability via breadth-first search on $G_{\text{structural}}$ scaled by QoS contracts (**FaultInjector**), entirely separate from the stochastic queuing dynamics of I_{dyn} (**SimPy**). The moderate cross-oracle agreement ($\rho = 0.627$) therefore represents true convergent validity between two independent formulations of failure impact, confirming that queuing introduces authentic behavioral divergence while verifying that SaG’s learned predictors function as fast structural surrogates for topological cascade simulation rather than detectors of unmodeled runtime queue dynamics (Section 8.3).

7.3.3. Node-Type Stratification and Attention

One result governs how every other number in this paper is read. Measured against $I_{\text{comp}}(v)$ over the twelve scenarios of the corpus, stratified RM rank correlations are $\rho = 0.597$ (Application), 0.317 (Broker), and 0.138 (Node), while pooling all types gives $\rho = 0.217$ — less than two fifths of the Application figure it is supposed to summarise. This is why we report every evaluation on a single stratum, and why pooled critical-set figures should be read as inflated wherever they appear. We do not claim the stronger form of the effect: pooled correlation is not below *every* stratum it aggregates, sitting above Node (0.138), so this is an aggregation bias rather than a strict Simpson reversal. An earlier version of this section reported the strict condition as holding; that figure was computed over an eight-scenario suite belonging to a companion study, which omitted five of this corpus’s scenarios and included a regression fixture, and the condition does not survive on the corpus this paper evaluates. A global sensitivity sweep of I_{comp} ’s four severity weights ($N = 1,000$ Dirichlet draws; Supplementary Section S1.2) shows Application correlation exceeding pooled on *all* draws ($\rho \in [0.441, 0.599]$ vs. $[0.096, 0.351]$), while the strict condition holds on only 13.9% of the simplex. The stratification argument rests on the former, which is weight-invariant. Evaluation of a rule-based anti-pattern catalog on the same benchmark indicates that the catalog flags 93.4% of scored components and therefore does not discriminate, so critical-set identification is delegated to the continuous rankers of Sections 7.1–7.2.

Aggregated by relation type over the ATM case study, first-layer mean HGT attention orders **USES** into libraries (0.227) above publish–subscribe channels (0.163–0.176). However, the spread across all seven relation types is narrow (0.15–0.23) and driven substantially by destination in-degree artifacts. Typed attention remains active across relation types without establishing a statistically distinct ordering.

7.4. RQ4: Real-World Distributed Architecture Validation

*RQ4 Summary (Real-World Transfer):. Full-population transfer ($\rho = 0.792$) is inflated by separating inert components from active ones: restricted to components that actually propagate failures, every predictor loses roughly 0.5–0.6 of correlation. HGT-QoS is the only one that stays positive ($\rho_{>0} = +0.281$); all three training-free baselines turn negative (-0.055 RM, -0.083 **Topo**, -0.092 **Topo-QoS**), so the collapse is a property of the active-stratum task rather than of relational learning. We therefore report RQ4 as a qualified positive: the typed model is the only predictor to retain any active-stratum signal on transcribed architectures, but $+0.281$ is too weak to establish transfer, and it still inverts on the synchronous Train-Ticket call tree.*

We evaluated the framework on five open-source distributed systems transcribed from public repositories: Online Boutique, Train-Ticket, Home Assistant, Autoware Universe (ROS 2), and EdgeX Foundry. All carry labels from the same simulation oracles used throughout, testing out-of-family topological transfer rather than agreement with field incident telemetry.

Two evaluations are conducted. The first is an exploratory evaluation of the closed-form explanation layer $Q(v)$ against the composite oracle $I_{\text{comp}}(v)$ ($\rho = 0.514$ – 0.800); as noted in Section 4.3, I_{comp} ’s four severity weights are unswept heuristics. The second, Section 7.4.1, evaluates learned predictors zero-shot against $I^*(v)$ (where RM achieves a lower mean $\rho = 0.516$). **Topo-QoS** is scored here on the flow projection, for the reason given in Section 8.4. Standard middleware default contract profiles (e.g., ROS 2 Best-Effort/Reliable, MQTT QoS 0/1) were applied uniformly across edges lacking explicit manifests to ensure identical graph representations across baselines.

7.4.1. Zero-Shot Transfer of the Learned Model

To assess generalization outside the generator, HGT-QoS was trained on all twelve synthetic scenarios and evaluated zero-shot across five open-source systems versus $I^*(v)$ (five seeds; evaluated in an exploratory 2-layer, 150-epoch configuration with within-graph rank normalization; no open-source graphs contributed training gradients or checkpoint selection). Table 11 reports the transfer performance.

Table 11: Zero-shot out-of-family topological transfer to five open-source systems, scored against $I^*(v)$ on the Application population (V_{app}), reporting both Spearman rank correlation (ρ) and critical-set identification ($F_1@K$ at $K = \text{round}(0.20 \cdot |V_{\text{app}}|)$). HGT-QoS trains on all twelve synthetic scenarios ($\pm$ = spread over five seeds); RM, Topo, and Topo-QoS are training-free and are evaluated on identical labels and components. Where open-source systems lack declared QoS manifests, default middleware contract profiles (e.g., ROS 2 Best-Effort/Reliable, MQTT QoS 0/1) were applied uniformly across edges. $\rho_{>0}$ restricts rank correlation to the $n_{>0}$ actively propagating components (Section 7.1.2). Bold indicates the best score per row.

Real-World Architecture	$ V_{\text{app}} $	$n_{>0}$	RM		Topo		Topo-QoS		HGT-QoS		
			ρ	F_1	ρ	F_1	ρ	F_1	ρ (Full)	$\rho_{>0}$ (Active)	$F_1@K$
Cloud Microservices Mesh	22	8	0.777	0.250	0.891	0.250	0.888	0.250	0.769 ± 0.054	+0.031	0.500 ± 0.158
Train-Ticket Booking Mesh	41	14	0.713	0.625	0.528	0.625	0.541	0.625	0.775 ± 0.016	-0.173	0.400 ± 0.050
Autoware.universe (ROS 2)	32	19	0.357	0.167	0.307	0.167	0.378	0.167	0.745 ± 0.070	+0.516	0.667 ± 0.149
EdgeX Foundry (Industrial IoT)	22	10	0.470	0.000	0.534	0.000	0.534	0.000	0.772 ± 0.055	+0.262	0.500 ± 0.000
Home Assistant (Smart Home)	24	17	0.265	0.200	0.297	0.200	0.289	0.200	0.902 ± 0.020	+0.770	0.600 ± 0.000
Mean	—	—	0.516	0.248	0.511	0.248	0.526	0.248	0.792	+0.281	0.533

Table 12: Real-world active stratum: means over the five open-source systems on the full Application population (ρ) and restricted to components with strictly positive ground-truth impact ($\rho_{>0}$), the counterpart of Table 9 on transcribed rather than generated architectures. All four predictors are scored against identical labels, populations and node sets from a single run, so the columns are commensurable. Δ is $\rho_{>0} - \rho$; a retention ratio is not reported because three of the four active-stratum means are negative, which makes a percentage of the full-population figure uninterpretable.

Predictor	ρ (full)	$\rho_{>0}$ (active)	Δ
RM / $Q(v)$	0.516	-0.055	-0.571
Topo	0.511	-0.083	-0.595
Topo-QoS	0.526	-0.092	-0.618
HGT-QoS	0.792	0.281	-0.511

Key Insights for Real-World Transfer:

- The full-population figure is not a ranking result.** On all Applications, HGT-QoS reaches $\rho = 0.792$ vs. 0.511 (Topo), 0.526 (Topo-QoS), and 0.516 (RM), leading on 4/5 systems. However, between 29% (Home Assistant) and 66% (Train-Ticket) of Applications carry zero simulated impact; full correlation heavily rewards separating inert from active components rather than ranking active ones. Table 12 therefore reports the active stratum for every predictor, not only the learned one.
- On the active stratum every predictor collapses, and only the typed model survives it.** Restricting to components with strictly positive impact costs HGT-QoS 0.511 of correlation, leaving $\rho_{>0} = +0.281$; it costs the training-free baselines more than they have, leaving all three negative (-0.055 RM, -0.083 Topo, -0.092 Topo-QoS; Table 12). The pub-sub systems carry the learned model’s margin (+0.770 Home Assistant, +0.516 Autoware, +0.262 EdgeX); Cloud Microservices is now indistinguishable from chance (+0.031, $n = 8$) and Train-Ticket still inverts (-0.173, $n = 14$). **We therefore report RQ4 as a qualified positive:** relational learning is the only approach here that retains active-stratum signal at all, but the magnitude does not establish transfer, and the synchronous inversion persists.
- The QoS-weighted baseline wins on the full population of one system, and loses the stratum that matters.** Topo-QoS reaches $\rho = 0.888$ on Cloud Microservices against the learned model’s 0.769. On that same system’s active components it scores -0.072 while HGT-QoS scores +0.031: the baseline’s full-population lead there comes from separating inert components, which is precisely the effect the active stratum removes.
- Critical-set identification shows higher top- K overlap in edge cases.** On operational triage ($F_1@K$), HGT-QoS averages 0.533 across the five benchmarks compared to 0.248 for structural baselines. Given the small K (4–8 components) and the cross-seed Jaccard instability, these metric differences

should be viewed descriptively. F_1 does highlight structural blind spots: on EdgeX Foundry, structural baselines score $F_1 = 0.000$ ($\rho \approx 0.47\text{--}0.53$), while HGT-QoS identifies 2 of the top 4 components ($F_1 = 0.500$). In EdgeX’s IoT topology, symmetric star connections from peripheral adapters to brokers produce identical betweenness ties that collapse triage at $K = 20\%$ ($K = 4$), whereas relational attention differentiates components via message flow features. Similarly, on Autoware.universe, HGT-QoS captures more top- K components (0.667 vs. 0.167), and on Cloud Microservices it achieves $F_1 = 0.500$ vs. 0.250.

5. Architectural Ingestion Boundary. The active-stratum inversion on microservices reveals a fundamental domain mismatch: synchronous RPC architectures propagate chain failures backward along invocation trees via timeouts and thread starvation [48], whereas asynchronous pub-sub architectures cascade forward via queue saturation and topic starvation. Because HGT-QoS was trained on pub-sub semantics, its directional bias inverts on synchronous call trees. A strict ingestion boundary is therefore required: SaG’s learned GNN pipeline applies to asynchronous and event-driven architectures (ROS 2, Kafka, DDS, MQTT), while call-graph analyzers should be used for synchronous RPC/REST microservice meshes. We do not recommend the closed-form baselines as the substitute there: Topo-QoS leads on Cloud Microservices only on the full population ($\rho = 0.888$) and is itself negative on that system’s active components (-0.072), so its advantage lies in identifying inert services rather than in ranking the ones that propagate.

7.5. RQ5: Analysis Cost and Its Comparison Against Simulation

RQ5 Summary (Analysis vs. Simulation Cost):. While the neural forward pass executes in milliseconds (56 ms for 2,000 nodes), cold deterministic feature extraction requires up to 239.3 s, dominated by the $O(|V|^2 + |V||E|)$ Connectivity Degradation Index. Cold static gating is thus roughly eighteen times slower than discrete-event simulation (79.3 s vs. 4.5 s on the 520-node Enterprise mesh), refuting the assumption that static analysis is inherently cheaper than simulation without incremental graph caching.

RQ5 quantifies computing overhead and sustainability during CI/CD evaluation, with per-stage latencies reported in Table 13:

Table 13: Per-stage latency of the inference pipeline across scaling graph sizes (CPU, median of 3 runs).

$ V $	$ E $	Analyze (s)	Graph $\rightarrow$ Tensor (s)	HGT Forward (ms)	Analyze : Forward
249	1,127	1.74	0.010	26.5	66×
499	2,402	8.32	0.022	16.4	509×
999	6,422	44.54	0.056	21.1	2,108×
1,998	19,301	239.34	0.157	56.2	4,259×

The neural model constitutes the least expensive stage, whereas the deterministic model incurs considerable computational cost. For 2,000 components, the HGT forward pass requires 56 ms, compared to 239 s for deterministic structural analysis, yielding a ratio of 4,259×. This ratio reflects pipeline cost rather than the total cost of architectural evaluation. Indices 0–17 of each node feature vector (Section 3.4)—including betweenness, closeness, reverse PageRank, articulation, and bridge scores—are generated during this analysis stage, making the forward pass dependent on its completion. End-to-end evaluation of an unseen 2,000-component architecture takes about four minutes, with the learned model accounting for only 0.02% of that time; the 56 ms figure represents the marginal cost of re-scoring an already-analyzed graph. Across the twelve-scenario corpus, the complete gate (structural analysis plus 18 anti-pattern detectors) executes in 0.16–79.3 s, with the upper bound corresponding to the 520-component Enterprise mesh.

One metric dominates cost, and it grew. Measured cost now tracks the stage’s $O(|V|^2 + |V||E|)$ bound closely: from 249 to 1,998 components wall-clock rises 138× against a 137× growth in $|V||E|$. The dominant term is the Connectivity Degradation Index, which is computed for every node in the main connected component rather than for articulation points alone. That choice is deliberate and is a correctness requirement rather than an oversight: gating CDI to articulation points leaves it identically zero for every node whose removal does not literally disconnect the graph, which drives $A(v)$ to a near-constant in the

redundant multi-publisher topologies this system targets. The cost is the price of a non-degenerate Availability score, and we report it rather than the cheaper gated variant we could have measured.

7.5.1. The Gate Is Not Cheaper Than the Simulation It Replaces

The framing that motivated this analysis — static gating as a low-cost substitute for dynamic simulation — does not survive measurement against our own oracle. Timing the cascade reachability labeling sweep (five seeds, node types Application/Broker/Library, the full ground-truth run) over all twelve scenarios, on the same machine, at the same commit and in the same measurement session as the gate, gives 0.08–4.49 s per scenario against 0.16–79.3 s for the analysis gate. Both maxima occur in the 520-component Enterprise mesh, so the largest scenario compares 4.5 s of simulation against 79.3 s of static analysis: the gate costs roughly eighteen times as much as the simulation it is meant to replace. These are wall-clock figures on one commodity CPU and should be read to one significant figure. Across sessions on identical work the gate maximum has ranged over 77–83 s and the oracle maximum over 4.5–4.8 s, so the ratio is stable to about ± 1 ; two independently paired sessions give $16.7\times$ and $17.7\times$. Two earlier pairings are superseded rather than reconciled: one reported $11.5\times$ from an oracle half carrying no provenance stamp, so neither the machine nor the commit behind it can be recovered, and one paired a twelve-scenario oracle against a gate measured over a companion study’s eight-scenario suite, whose lower bound of 0.04 s came from a regression fixture outside this corpus.

This finding refutes the assumption that static analysis is computationally cheaper than in-process simulation: breadth-first cascade traversal is simpler than computing all-pairs connectivity degradation ($O(|V|^2 + |V||E|)$). However, in practical continuous integration workflows, static SSA provides a key deployment trade-off. First, it scores components (e.g., shared libraries, hosts) and dependency edges that node-level simulation passes do not evaluate. Second, deterministic graph metrics can be incrementally cached across git commits, recomputing only the k -hop neighborhood touched by an architectural pull request. We clarify that the benchmark timings in Table 13 reflect full from-scratch recomputation without caching. Under an incremental caching design, GNN scoring would execute in 56 ms for unchanged subgraphs, but such caching is not currently implemented in our pipeline, whereas repeating full simulation sweeps requires re-running stochastic traversals globally. Without such caching, direct simulation is strictly faster.

8. Discussion, Threats to Validity, and Limitations

8.1. Discussion and Practical Consequences

When to use Topo-QoS, and when to use HGT-QoS. The findings do not support an unequivocal recommendation of the learned model over the closed-form alternative; we present empirical trade-offs directly.

- **Training-free ranking (Topo-QoS) vs. Direct Simulation.** Topo-QoS eliminates training, checkpoint storage, and tuning, achieving $\rho = 0.553$ zero-shot across twelve synthetic architectures and 0.526 across five open-source systems. Learned models do not yield statistically significant ranking gains ($+0.085$, $p = 0.151$, 95% CI spanning zero; and 0.888 vs. 0.769 favoring baseline on Cloud Microservices’ full population). Thus, Topo-QoS serves as the zero-maintenance default for scalar ranking in resource-constrained CI pipelines. Alternatively, where queue parameters exist and graphs are small-to-medium ($N < 500$), direct simulation (0.14–7.2 s) offers exact ground-truth traces without surrogate approximation.
- **One learned mechanism, not two (GAT-N-QoS or HGT).** Relation typing or QoS edge channels individually yield substantial improvements over unweighted baselines ($+0.234$ and $+0.287$, Holm-corrected $p \leq 0.0015$); combining both provides minimal additional benefit. Untyped QoS-weighted GNNs offer a superior efficiency trade-off, achieving $\rho = 0.604$ with 28,168 parameters vs. HGT-QoS’s 0.638 with 434,620 parameters. Unweighted homogeneous models (GAT-N, $\rho = 0.317$) underperform heuristics, indicating GNNs lacking relational signals fail on this task.

- **Capabilities without a closed-form counterpart.** The typed model exposes one unique diagnostic: per-relation attention (Section 7.3.3). However, measured spread across relation types is narrow (0.15–0.23), governed by destination in-degree, and unstable under regeneration; it establishes that typed attention is active, not that it is diagnostically reliable. We decline to advance this as a reason to adopt typing on the present evidence.
- **Ingestion boundary.** Deployment must condition on communication synchrony (Table 14): on asynchronous pub-sub systems (ROS 2, DDS, MQTT, Kafka) HGT-QoS transfers positively on active components, whereas on the synchronous Train-Ticket call tree it inverts and should not be deployed — use static call-graph reachability tools instead. Topo-QoS is not the substitute: it is negative on the active components of both RPC meshes.

Architectural and Systemic Drivers of Graph Learning Success. Across seventeen architectures, graph neural network performance is governed by the five systemic factors in Table 14.

Table 14: Systemic drivers of graph-learning performance across the seventeen evaluated architectures, with the mechanism each reflects and the evidence for it on this corpus.

Factor	Mechanism	Evidence on this corpus
Communication synchrony	Directional bias must match physical failure flow: pub-sub starves forward, synchronous call trees time out backward [48].	Pub-sub $\rho_{>0} \in [+0.262, +0.770]$; RPC collapses to chance or below (+0.031 Microservices, -0.173 Train-Ticket), while every training-free baseline is negative on both.
Scale and diameter	Fixed 3-layer message passing covers the diameter of small graphs ($N < 150$) but localizes at $N \geq 500$.	520-node Enterprise: HGT-QoS 0.461 vs. Topo-QoS 0.795, where global shortest paths resolve bottlenecks 3-hop convolutions miss.
Topology and symmetry	Isomorphic 1-hop neighborhoods yield near-identical embeddings; dense irregular meshes do not.	Broker-hub ESB $\rho = 0.472$; on microservices and ATM, relational attention gains +0.210 to +0.229 over Topo-QoS.
Relational heterogeneity	Typing is essential where shared hosts or libraries induce blast radii, but substitutes for QoS once contracts are declared.	Untyped GNNs drop to $\rho = 0.317$ (Section 7.2).
Inert-node base rates	Zero-impact sinks inflate full-population correlation through trivial inertness filtering.	21%–52% of applications carry $I^*(v) = 0$; $\rho_{>0}/\rho \approx 51\%$ –56% (Section 7.1.2).

Dual-Engine Consensus Protocol. Because prediction dispersion $\hat{\sigma}$ correlates negatively with margin over Topo-QoS ($\rho_s = -0.126$, Section 7.2.1), automated fallback is withdrawn. Instead, SaG executes dual-prediction evaluating HGT-QoS and Topo-QoS concurrently, flagging unanimous top- K nodes and highlighting divergent rankings for review.

Role of the Explanation Layer. The RM profile ($Q(v)$, Section 5) provides standards-compliant architectural diagnostics under ISO/IEC 25010. While neural and centrality predictors govern triage priority, RM decomposes structural mechanics to inform refactoring (distinguishing single-point-of-failure Availability replication from cascade Fault Tolerance circuit breakers). However, elicited AHP weights rank worse than a uniform prior; pending user studies, we recommend the uniform prior.

8.2. Performance and Computational Sustainability Implications

What sustainability means for a pre-deployment gate. Green software engineering assesses energy across development, assurance, and execution [29, 88, 89, 90, 91, 30, 31]. While live chaos sweeps require cluster-hours across provisioned nodes, pre-deployment static analysis aims to eliminate cloud staging footprints. Direct physical energy measurement in joules requires dedicated hardware counters (RAPL, NVML) and profilers [30, 32, 33], which we outline as an instrumentation protocol for future testbeds.

Where the cost actually sits. We withdraw the efficiency claim: cold static analysis does not reduce CPU computation over simulation. Global connectivity degradation (79.3s on Enterprise) is roughly eighteen

times slower than cascade traversal (4.5 s) due to CDI’s $O(|V|^2 + |V||E|)$ cost. In CI, sustainability would be achieved via incremental graph caching: caching base metrics across commits and extracting features only for PR delta subgraphs would reduce latency to the sub-second forward pass (56 ms). Without caching, static analysis consumes more CPU time than the simulation it displaces.

8.3. Threats to Validity

Construct Validity. Ground-truth impact $I^*(v)$ is derived from simulation rather than live production outages. While I^* correlates with dynamic queue flow I_{dyn} ($\rho = 0.627$ against a 0.811–1.000 ceiling, Section 7.3.2), top- K Jaccard reaches only 0.27–0.37. Furthermore, $I^*(v)$ is recovered at $\rho = 0.965$ by topology-only relabeling, confirming the target captures reachability rather than runtime anomalies. Additionally, transcribing open-source architectures into multigraphs relied on standardized heuristics without multiple independent coders, introducing potential transcription subjectivity.

Internal Validity. Feature leakage is prevented by strict graph separation: predictors consume G_{analysis} , while simulation oracles traverse $G_{\text{structural}}$ (CI-asserted). Parity is maintained via matched datasets, depths, and early stopping. However, comparing HGT-QoS to GAT-N-QoS conflates typing with a $15.4\times$ capacity gap (434,620 vs. 28,168) and bidirectional passing (103,725 reverse-projection parameters in HGTConv), requiring registered control variants (GAT-N-C, GAT-N-QoS-C, HGT-QoS-U, Section 8.4) to isolate. Six active QoS dimensions govern profiles, with one reserved extension point.

External Validity. The evaluation covers twelve synthetic scenarios and five open-source systems. Zero-shot transfer to active components is weak for every predictor: HGT-QoS retains $\rho_{>0} = +0.281$ while all three training-free baselines turn negative (-0.055 to -0.092 , Table 12), and the learned model still inverts on a synchronous call tree (Section 7.4.1). The bound this places on external validity is therefore about the active stratum as a task, not about relational learning specifically. Microservices cascade backward along call trees via timeouts [48], whereas pub-sub cascades propagate forward via starvation. GNN directional biases must therefore be conditioned on communication synchrony. Scaling beyond 2,000 nodes requires incremental caching or mini-batching (GraphSAINT [92]).

Conclusion Validity. Distributions are evaluated using non-parametric correlations (Spearman ρ , Kendall τ), bootstrap intervals ($B = 2,000$), and Wilcoxon signed-rank tests. Analyses are stratified because pooling node types aggregates strata whose correlations differ by a factor of four (Node $\rho = 0.138$ to Application 0.597), leaving the pooled figure (0.217) at less than two fifths of the stratum this paper actually scores on. We do not claim a strict Simpson reversal on this corpus: pooled correlation exceeds the Node stratum. Because folds share training graphs, test observations exhibit positive dependence; nominal Wilcoxon p -values are optimistic and interpreted descriptively. Primary non-parametric evidence rests on 12-fold sign consistency, though sign tests over overlapping splits inherit positive dependence as well. Furthermore, synthetic graphs derive from a single generator family, bounding empirical generalizability.

Repeatability. Seed spread and run-to-run displacement are distinct quantities, and only the former is reported in Table 8. At fixed code and fixed seeds the pipeline is deterministic: two consecutive executions of the zero-shot transfer experiment reproduced every reported figure exactly, learned and training-free alike. Across a code revision, however, the two halves behave differently. Re-executing the in-distribution grid at identical seeds and against the identical corpus digest, after intervening changes to the codebase, reproduces all 120 training-free cells exactly (maximum $|\Delta\rho| = 0.0000$), while the 240 learned cells move by up to 0.93 at the level of an individual seed-cell and 0.21 at the level of a scenario mean (AV/HGT, $0.637 \rightarrow 0.845$) — a displacement exceeding the $+0.085$ primary contrast. The same revision moved the real-world active stratum itself, cutting $n_{>0}$ by between 26% and 56% per system, so $\rho_{>0}$ figures are not comparable across code revisions even though each is individually reproducible. The practical consequence is that a learned figure in this paper is pinned by the commit its artifact names and not merely by its seed, whereas a structural figure is pinned by the corpus alone; the provenance stamp carried by every artifact (Data Availability) is what makes that distinction checkable rather than assumed.

8.4. Limitations and Future Work

Correction of the Real-World Baseline. On raw multigraphs, betweenness vanishes for Application nodes; evaluating on DEPENDS_ON flow projections enables Topo-QoS across all five real-world systems (Table 11).

Explanation Layer Validation. SaG separates Availability from Fault Tolerance, but human studies have not yet validated actionability. Elicited AHP weights perform worse than a uniform prior at ranking (Section 7.3); user evaluations and mutation benchmarks are prioritized for future work.

Uncontrolled Confounds in Typing. Table 10 holds substrate, training set, depth, and early stopping constant, but parameter capacity (434,620 vs. 28,168) and reverse message directionality (103,725 reverse-relation parameters in HGTConv) remain unmatched. Because $I^*(v)$ measures downstream reachability, upstream visibility confers an advantage unrelated to typing; the +0.134 effect reflects the joint transition to HGT. Registered control variants (GAT-N-C, GAT-N-QoS-C, HGT-QoS-U) are prioritized to isolate these factors.

Absence of Live Incident Telemetry. Ground-truth simulation oracles ($I^*(v)$, I_{comp} , I_{dyn} , I_M) derive from simulation and static fault injection. While ensuring determinism and byte-identical regeneration, this does not replace post-mortem incident logs, distributed tracing (e.g., OpenTelemetry spans), or physical chaos experiments on live clusters. Validating topological cascades against production outages remains an essential prerequisite for industrial adoption.

Model Selection and Incremental Caching. Early stopping employs an inner validation split on the primary graph; held-out scenario validation is a prioritized extension. Prediction dispersion does not reliably signal out-of-distribution fallback (Section 7.2.1). Production deployment requires incremental graph caching over pull request diffs to amortize feature extraction.

Future Directions. Key extensions include: (1) modeling distributed LLM serving backbones (vLLM, DeepSpeed); (2) measuring hardware energy via RAPL/NVML to benchmark static gating against live sweeps in joules; and (3) advancing to prescriptive synthesis, generating automated pull requests with circuit breakers.

9. Conclusion

This study investigated the empirical viability and architectural boundaries of heterogeneous graph neural networks for pre-deployment dependability analysis in asynchronous distributed systems. Using Software-as-a-Graph (SaG) as an experimental framework transforming Architecture-as-Code manifests into typed multigraphs, we evaluated a relation-specific Heterogeneous Graph Transformer with Quality-of-Service edge encodings against unparameterized structural baselines and an interpretable ISO/IEC 25010/25019 Reliability–Maintainability attribution layer without runtime telemetry.

The primary empirical finding is that the framework’s two architectural mechanisms substitute rather than complement each other. Relation typing and 16-dimensional QoS edge encodings exhibit individual main effects under inductive distribution shift ($\Delta\rho = +0.134$ and $+0.187$, Holm-corrected $p = 0.0015$, with typing reflecting the joint transition to HGT). However, their interaction is -0.199 , remaining negative across all twelve folds ($p = 0.0005$): typing contributes $+0.234$ without QoS and $+0.035$ with it. Either mechanism alone recovers most of the full model’s performance; combining them yields minimal benefit, indicating both channels encode traversal semantics. The oracle supports this: topology-only relabeling recovers $I^*(v)$ ’s ordering at mean $\rho = 0.965$ without QoS terms.

This finding carries direct practical implications: teams requiring scalar rankings should consider the simpler untyped QoS-weighted baseline ($\rho = 0.604$ at 28,168 parameters vs. HGT-QoS’s 0.638 at 434,620, non-significant despite a $15.4\times$ capacity gap), as relation-specific attention is narrow and sensitive to degree (Sections 7.3.3 and 8.1). Boundary conditions are clear: learned ranking does not significantly outperform unparameterized QoS centrality ($+0.085$, $p = 0.151$), zero-shot transfer on active components is weak for every predictor and only the typed model stays positive ($+0.281$ vs. -0.055 to -0.092 , still inverting on a synchronous call tree), prediction dispersion fails as an OOD indicator, and elicited AHP weights rank poorer than a uniform prior.

SaG provides a reproducible multigraph formulation of publish-subscribe systems, a byte-identically regenerating corpus, and an evaluation of where architectural inductive biases compose. The pipeline incorporates a learned component three orders of magnitude faster than its deterministic counterpart, which is itself roughly eighteen times slower than simulation unless incremental caching is applied across commits. Validating whether such pipelines predict live production outages remains the central empirical frontier.

Declarations

CRedit Authorship Contribution Statement. **Ibrahim Onuralp Yigit:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Feza Buzluca:** Conceptualization, Methodology, Validation, Writing – review and editing, Supervision.

Declaration of Competing Interest. The authors declare no competing financial interests or personal relationships that could have influenced this work.

Funding. This research received no grant from public, commercial, or not-for-profit funding agencies.

Data Availability. The replication package (datasets, harnesses, checkpoints, scripts) is available on Zenodo under DOI 10.5281/zenodo.14922108 [93] with `uv/pip` environments. Synthetic datasets regenerate byte-identically. The deposit ships the artifacts backing every reported table as a single dated bundle (`SaG_JSS_Results_<stamp>/`), assembled by `reproduce/cut_results_bundle.py`: sixteen JSON artifacts plus the tables and figures rendered from them, and a `MANIFEST.json` recording, per file, a SHA-256 digest, the commit it was produced at, whether that working tree was modified, and the corpus digest the run describes. Five artifacts (`atm_scale_sweep_v3`, `loso_significance_v5`, `qos_label_ablation`, `threshold_sensitivity_v3`, `topic_weight_sensitivity_v3`) carry no provenance block, so for those the correspondence to this corpus is asserted by the bundle rather than proved by the artifact. The verification script (`reproduce/reconcile_manuscript.py`) runs standalone on a clean machine against the Zenodo deposit, verifying table quantities against JSON artifacts with git provenance. Prose definitions, equations, and sensitivity tables are verified manually.

Declaration of Generative AI. The authors used Anthropic’s Claude for typesetting assistance, taking full responsibility for all content. No generative AI was used to design experiments, analyze data, or generate results.

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